Deeply Inelastic Scattering Revisited

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Abstract

Due to its conceptual simplicity, Deeply Inelastic Scattering (DIS) often serves as a textbook example for Quantum Chromo Dynamics (QCD) and partonic structure of nucleons. However, it is this supposed simplicity, which makes DIS an ideal case for discussing many additional effects and corrections. In these lectures we discuss DIS within the domain of perturbative QCD covering a range of topics ranging from mathematical problems to kinematical corrections, and to practical challenges when attempting interpret experimental measurements. Although here we focus on the exemplary case of DIS, the discussed topcis are relevant to a larger class of processes. We also overview existing and future DIS measurements, and discuss their implementation into modern extractions of parton distribution functions (PDFs) which quantify the partonic structure of nucleons.

The discussed features are all implemented in the Yadism [1, 2] open source code, which has been used for PDF extraction from a wide range of the available world data on fully inclusive DIS.

Acknowledgements

First, I want to thank Werner Vogelsang, who not only introduced me into the world of pQCD, DIS, and PDFs, but who also gave me the opportunity to explore all the glorious details, which we happily neglect in the rest. It turns out, recently, there is a renewed interest in my PhD thesis [3] which discusses a fairly complicated DIS process.

Second, I want to thank the members of the NNPDF collaboration and in particular Alessandro Candido, Giacomo Magni, and Stefano Forte, with whom I sorted out in long discussions even more details of DIS. Together, we have eventually brought all our experience and expertise into Yadism [1], which, hopefully, serves the community well in the coming years.

Third, I want to thank Ilkka Helenius and Hannu Paukkunen, who have supported me in various ways, gave me a lot freedom to work on my projects, and have shown me how to apply my knowledge to other, new fields.

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Preface

Here is the usual list of disclaimers:

- This is a highly personal selection of topics, which we consider relevant enough for presenting in this occasion and by no means a review of DIS, not even fully inclusive DIS in the collinear framework using pQCD, which is the only topic here.
- Since this is not a review also the list of reference is not exhaustive, but rather a starting point and the reader should consult the mentioned works for further works.
- This work may contain errors, such as typographical or grammatical, but hopefully no physical, mistakes. Their distribution and the overall normalization is currently unknown, but we hope the former is highly localized and the latter small.

Contents

List of Corrections

Chapter 1.

The basics

This chapter serves to bring everybody to the same starting point, by quickly recalling the most important features of the underlying theory and the required practical tools. We refrain here from giving a detailed introduction, which instead can be found in any standard textbook [4–8], and focus instead on practical problems, which we then discuss in detail in Chapter 2.

1.1. Introduction

We aim in this course to make a meaningful comparison between the theoretical predictions and the experimental measurement of a fully inclusive Deeply Inelastic Scattering (DIS) cross section. Instead of an abstract, theoretical discussion about DIS, which can be found in a standard lecture on particle physics, we're looking here to make a connection with real-life physics.

1.1.1. Motivation and non-goals

The main focus is on Parton Distribution Functions (PDFs), for which DIS plays a major role. In fact, DIS was, is, and will be a very important constrain in determining PDFs - only recently, e.g. starting from NNPDF4.0 [9], LHC data starts to dominate in PDF fits. Stressing the importance and relevance of PDFs for any kind of perturbative QCD (pQCD) application and to understand the inner structure of hadrons is left to the literature [10–12]. In turn, DIS itself can be used to answer unresolved questions about PDFs, e.g. whether the proton has an intrinsic charm component [13, 14].

In order to obtain a more realistic DIS treatment, we review general pQCD features and exemplify them using DIS. In fact, fully inclusive DIS comes in very handy, since here the effects mostly appear in their simplest form and when considering hadronic collisions, e.g. at the LHC, the strategies are mostly the same just more involved. We keep a phenomenological point of view with a specific focus on the numerical implementation as, eventually, we want to compare to the experimental measurements. We refrain from giving detailed proofs in this course and point to the standard textbooks [4–8] for that. Instead, we consider different features as given and focus on their practical implications. However, a major complication in this course is the interconnection between almost all features and while we discuss them here in turn, we highlight also how they are intertwined with each other.

In recent years (and much more so hopefully in the coming years), DIS has become a major research topic again thanks to the advent of the Electron Ion Collider (EIC) [15]. While the planned physics program of the EIC goes far beyond standard fully inclusive DIS, the simplest case discussed in this work can serve as an introduction into more complicated topics.

The structure of this course follows in large parts the implementation of Yadism (Yet Another DIS module) [1, 2], which is developed and used by the NNPDF collaboration in their PDF determinations. The online documentation, which is available at

`https://yadism.readthedocs.io/` ,

may serve as additional, more technical introduction into the topic. Note that Yadism is an open-source project to which everybody can contribute[1].

Since DIS is a very wide field, it is worth spelling out explicitly some non-goals:

- We only consider pQCD in the collinear framework here. However, because DIS is such a powerful experiment, which offers a clean access to the inner structure of the protons it can be studied using other frameworks as well, such as Color Glass Condensates [16, 17] → ask H. Mäntysaari or T. Lappi
- We only consider fully inclusive cross sections here (to be defined in Section 1.2 and with a tiny exception to be discussed in Section 2.5). In particular, we do not consider any more complicated final states, such as, e.g., single jet or di-jet production [18, 19].

[1] e.g. if some documentation can be improved

- We only consider pQCD in the collinear framework here. While, Monte Carlo Event generators, such as, e.g., Pythia [20], are rooted in collinear pQCD they eventually go much beyond and we do not consider their details here [19] → ask I. Helenius.
- We only consider DIS here. In particular, we do not consider photo-production or elastic or diffractive cross sections. We define the necessary kinematics for the distinction below in Section 1.1.3.
- We compute the theoretical predictions here. So while we try to keep the connection to the experimental side close, we do not discuss any specific details about how to measure cross sections.

1.1.2. Requirements

We need some basic knowledge from particle physics and, in particular, you need to be familiar with the Standard Model overview shown in Fig. 1.1. We discuss in this course each and everyone of these particles, with the exception of the three heaviest in each category: the heavies quark, the top, the heaviest lepton, the tau, and the heaviest boson, the Higgs. They work, in principle, mostly like their lighter brothers and sisters, but in practice are basically always neglected. We give a more thorough justification to discard them when we get to the actual numbers. All other particles explicitly appear in this course and you need to know their properties.

Conclusion 1.1: Summary forces, particles, and charges

- The carrier of the strong force is the gluon
- Quarks have a strong charge, leptons don't
- The carriers of the electro-weak force are the photon, the Z boson, and the W boson
- Quarks and charged leptons have electro-magnetic charges
- Quarks and leptons have weak charges

We also need some basic knowledge from Quantum Field Theory (QFT). To start the program we require a Lagrangian and while we focus mostly on $\mathcal{L}_{\text{QCD}}$ we need also the interaction with the electro-weak bosons, so, strictly speaking, we need the full standard model $\mathcal{L}_{\text{SM}}$. Based on that we generate a set of Feynman rules and from that point

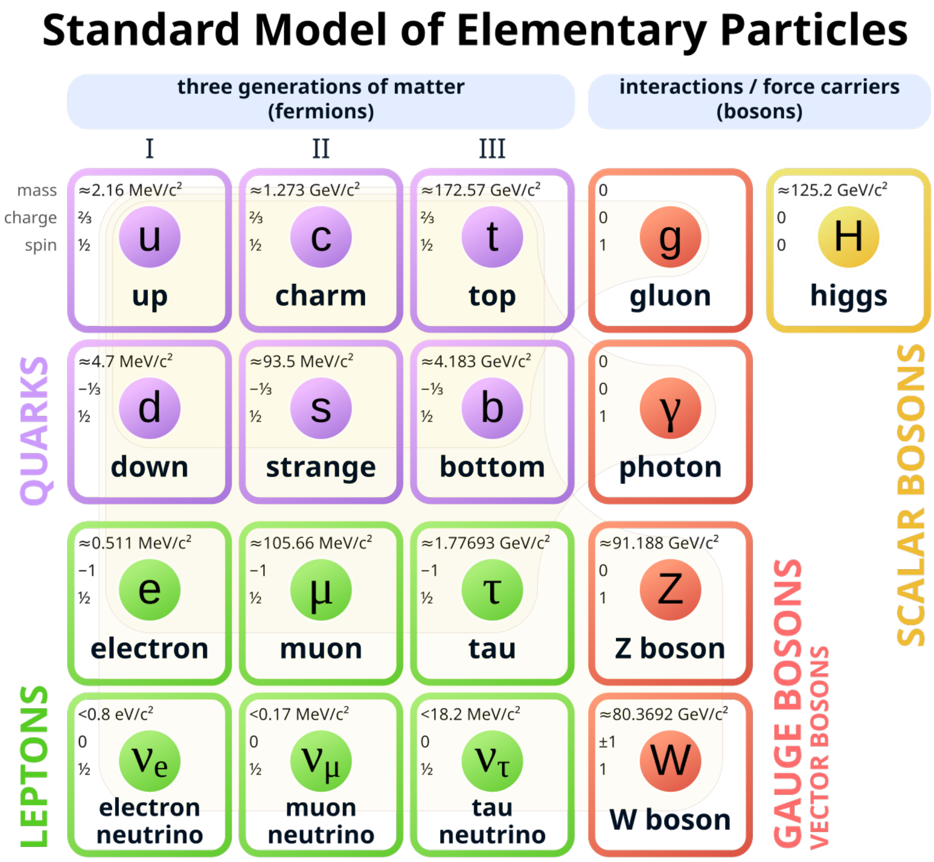

Figure 1.1.: The Standard Model of Elementary Particles - taken from [21]

onwards we are only talking in the language of diagrams. As a first step we perform the renormalization of all couplings and fields to remove any ultra-violet (UV) divergencies.

Conclusion 1.2: pQCD

- To compute any perturbative object, we sum over all relevant diagrams and then multiply them with their complex conjugated counterpart.
- For measurable cross sections we need to integrate over all internal and external particles.

1.1.3. The very basics of DIS

Conclusion 1.3: DIS in words

fully inclusive DIS = fully inclusive deeply inelastic scattering

What does that actually mean? We begin from the back:

- "scattering" means, of course, we are scattering two particles with each other. In this particular case, we are scattering a lepton off a hadron and to make things simple in the beginning we consider an electron is scattered off a proton.
- "inelastic" means we are breaking the target, the proton, apart; it shatters into many pieces.
- "deeply" is an adverb[2] to inelastic and it means we are looking "deep" into the inside of the proton. And in order to resolve small things, as usual, we need an highly energetic probe.
- "fully inclusive" means we are observing a single particle in the final state, which is the electron, but nothing else.

Thus, we can write the same information again with a formula

Conclusion 1.4: DIS in a formula

fully inclusive DIS = $e^-(k) + p(P) \to e^-(k') + X$ with $-(k-k')^2 \gg \Lambda^2_{\mathrm{QCD}}$

[2] thus it is "deeply inelastic" or "deep-inelastic", but not a list of adjectives "deep inelastic"

where we have assigned the four-momenta k, P, k' to the incoming electron, the incoming proton and the outgoing electron respectively. As usual X marks the inclusiveness, i.e. we don't observe anything else apart from the electron. By requiring a large momentum-transfer between the incoming and the outgoing electron, i.e. their difference four-momentum vector has a large magnitude, we probe the proton with a highly energetic parton (which we can think of as photon for now).

Next, we need a way to actually compute theory predictions for the cross sections, which you can measure in an experiment. We use the factorization theorem [22],

Conclusion 1.5: Factorization Theorem

$$\sigma(x,Q^2) = \sum_j \int_x^1 \frac{\mathrm{d}z}{z} C_j(z,Q^2) f_j(x/z,Q^2) + \mathcal{O}\left(\frac{\Lambda^2_{\mathrm{QCD}}}{Q^2}\right) \tag{1.1}$$

$$= \sum_j \left(C_j(Q^2) \otimes f_j(Q^2)\right)(x) + \mathcal{O}\left(\frac{\Lambda^2_{\mathrm{QCD}}}{Q^2}\right) \tag{1.2}$$

$$= \left(\mathbf{C}^T(Q^2) \otimes \mathbf{f}(Q^2)\right)(x) + \mathcal{O}\left(\frac{\Lambda^2_{\mathrm{QCD}}}{Q^2}\right), \tag{1.3}$$

which states how to compute a cross section $\sigma(x,Q^2)$ from coefficient function $\mathbf{C}(z,Q^2)$, computable in pQCD, and a universal PDF $\mathbf{f}(\xi,Q^2)$. In the remaining part of the course we discuss, enhance and correct this formula in many different ways and basically challenge every aspect. We define each and every variable and function in Section 1.2, so we can focus for the moment on three other important aspects: universality, measurability, and notation.

First, the crucial, physical aspect about Eq. (1.1) is universality: for any cross section σ measured at any x and Q^2 there is exactly one PDF $\mathbf{f}(\xi,Q^2)$. Instead, the coefficient functions $\mathbf{C}(z,Q^2)$ may depend on the definition of the cross section, but they are computable from first principles. This implies the predictive power of our framework: if $\mathbf{C}$ is fixed and $\mathbf{f}$ is universal, we can measure σ in one set of experiments and then predict the outcome of another set of measurements σ'.

Second, it is important to keep in mind that Eq. (1.1) contains only a single measurable quantity: the cross section. Instead, coefficient functions $\mathbf{C}$, PDFs $\mathbf{f}$ and also the strong coupling α_s are not directly measurable in a real experiment but have to be constructed consistently within the given framework of pQCD. For example, while cross sections have to be positive by definition, this is less trivial for PDFs [23–25].

Third, a more practical aspect of Eqs. (1.1) to (1.3) is the short-hand notation we introduce there. While Eq. (1.1) is the most canonical form of the factorization theorem, which can be found like this in most textbooks, we introduce in Eq. (1.2) the multiplicative convolution symbol $\otimes$. Note that we have suppressed the variable over which the convolution is performed and recall that convolution is associative, i.e. $(f \otimes g) \otimes h = f \otimes (g \otimes h)$. Finally, in Eq. (1.3) we introduce the vector notation, where we denote objects, which have a non-trivial dependence on the parton flavor j, with bold font. We use vector notation and the index notation interchangeably depending on the context. Equation (1.3) looks nice and compact, but it worth recalling that it already hides a lot of details: specifically, $C_j(z, Q^2)$ is an object which depends on three different variables.

To Leading Order (LO) accuracy, the coefficient functions $\mathbf{C}$ greatly simplify and we find

$$\sigma(x, Q^2) = \sum_j e_{q_j}^2 f_j(x, Q^2)\,, \tag{1.4}$$

i.e. the cross section is given by the sum of the quark PDFs weighted by their electrical charges squared. Note that here the variables, which are used for the experimentally measured cross section on the left-hand-side, are also used as arguments of the PDFs on the right-hand-side. This is a pure coincidence only valid in the simplest case at LO and two historical comments can be made here. First, when studying (older) literature this equivalence can lead to confusion as, e.g., PDFs have been also called "structure functions" at the time, or what exactly we mean when talking about a general variable "x". Second, in the so-called "parton model", which was popular before QCD was invented [26], the index j just referred to a "parton" with some properties. Instead, within QCD we now know exactly, what the object in question is: a quark, which has a strict definition in QFT with well-defined properties and quantum numbers. For simplicity, we still call an object, which can be either a quark or a gluon "parton".

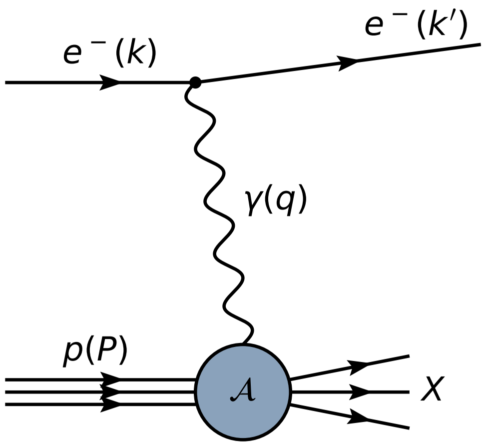

Figure 1.2.: Fully inclusive DIS represented as Feynman diagram

1.2. Setting the stage

In the following section we collect and define all ingredients needed for Eq. (1.1). As already said, DIS is given by

$$e^-(k) + p(P) \to e^-(k') + X \tag{1.5}$$

where the four-momenta k, P, k' belong to the incoming electron, the incoming proton and the outgoing electron respectively as shown in Fig. 1.2. This allows us, first, to define four momentum of the exchanged photon

$$q = k - k' \tag{1.6}$$

and from there the set of four common variables:

$$s_l = (k+P)^2 = \frac{Q^2}{xy}\,, \tag{1.7}$$

$$Q^2 = -q^2\,, \tag{1.8}$$

$$x = \frac{Q^2}{2q\cdot P}\,, \tag{1.9}$$

$$y = \frac{q\cdot P}{k\cdot P}\,. \tag{1.10}$$

We refer to Q^2 as virtuality (of the exchanged boson), to x as Bjorken-x (named after James Bjorken and often named x_{Bj}), and to y as inelasticity. Note that Eq. (1.7) gives an explicit relation among these, thus we have only three independent and one dependent variable. In practice, the invariant mass of the electron-proton system s_l is fixed by the experimental setup, i.e. by the energy of the colliding particles. Also the measurable range of y is in practice limited by the experimental geometry. DIS cross sections are typically provided as double-differential cross sections in x and y. Finally, it is convenient to define the mass squared of the system X recoiling against the scatter electron,

$$W^2 = (P+q)^2 = Q^2 \left(\frac{1}{x} - 1\right) = s_h\,, \tag{1.11}$$

which also corresponds to the invariant mass of the photon-proton system s_h.

Conclusion 1.6: DIS requirements

We require $Q^2 \gg \Lambda^2_{\text{QCD}}$ (deep) and $W^2 \gg \Lambda^2_{\text{QCD}}$ (inelastic) to call a lepton-hadron scattering DIS.

We can decompose the (double-differential) cross section into a leptonic interaction and a hadronic interaction, yielding each a tensor in Minkowski space, $L_{\mu\nu}$ and $W_{\mu\nu}$ respectively. Thus, we can write

$$\frac{\mathrm{d}^2\sigma}{\mathrm{d}x\mathrm{d}y} = \frac{2\pi y \alpha^2_{\text{em}}}{Q^4} L_{\mu\nu} W^{\mu\nu} \tag{1.12}$$

where we have introduced the electro-magnetic coupling α_{em}, which appears once on the leptonic side and once on the hadronic side. We can express the leptonic tensor using plain Quantum Electrodynamics (QED) and provide a decomposition of the hadronic tensor in terms of scalar functions. Together, we obtain

$$\frac{\mathrm{d}^2\sigma}{\mathrm{d}x\mathrm{d}y} = \frac{4\pi\alpha^2_{\text{em}}}{xyQ^2}\left(\left(1 - y + \frac{y^2 x}{2}\right) F_2(x,Q^2) - \frac{y^2 x}{2} F_L(x,Q^2)\right) \tag{1.13}$$

where we have introduced the two common scalar functions, now called structure functions F_2 and F_L, which are often measured in experiments. It is convenient to also define a third, linearly dependent structure function via

$$2xF_1(x,Q^2) = F_2(x,Q^2) - F_L(x,Q^2)\,. \tag{1.14}$$

Keep in mind that the discussion so far is a major simplification of DIS and in Section 2.1 we reintroduce the main complexity.

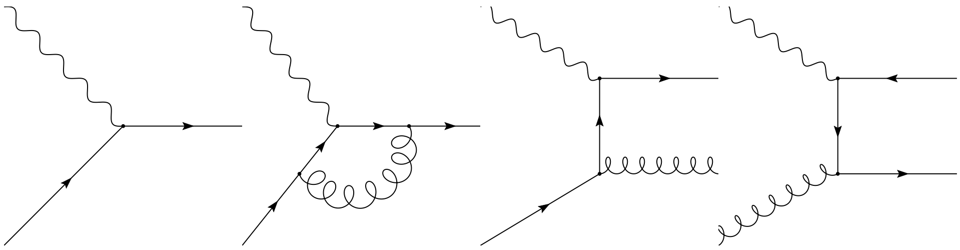

Figure 1.3.: Representative LO and NLO DIS Feynman diagrams. While at LO there is a single diagram (a), starting from NLO there are several type of diagrams: virtual corrections (b), radiative corrections (c), and the gluon as initial state parton (d). Note that the list is not exhaustive.

Let us continue to collect the remaining ingredient for Eq. (1.1):

- The sum over partons j includes quarks and gluons and we refine in Section 2.5 how large the sum actually is.
- The parton momentum fraction $\xi = x/z$ is the fraction of the four momentum the parton carries with respect to its parent proton.
- The probability of such a configuration to occur is given by the PDF, i.e. $f_j(\xi, Q^2)\mathrm{d}\xi$ is the probability to find a parton j with momentum ξP inside a proton with momentum P in an experiment performed at the scale Q^2.
- The integration over z sums all configuration to generate the final state configuration.

Finally, the coefficient functions $\mathbf{C}(z, Q^2)$ are computable order-by-order in pQCD and thus can be decomposed as

$$\mathbf{C}(z, Q^2) = \sum_{k=0} \alpha_s^k(Q^2)\mathbf{C}^{(k)}(z)\,. \tag{1.15}$$

Note that the Q^2 dependency of $\mathbf{C}(z, Q^2)$ is fully captured by the strong coupling $\alpha_s(Q^2)$.

1.3. Leading order and next-to-leading order

At LO accuracy, DIS is trivial, Fig. 1.3a, and we find

$$C_q^{(0)}(z) = C_{\bar{q}}^{(0)}(z) = e_q^2\delta(1-z)\ \forall q\,, \quad C_g^{(0)}(z) = 0 \tag{1.16}$$

where q represents any quark and g represent the gluon. Inserting Eq. (1.16) into Eq. (1.1) yields Eq. (1.4).

Starting from next-to-leading order (NLO), the diagrams become more complicated and we encounter new features:

- We have to perform renormalization at the requested order to remove all UV divergencies.
- We encounter loop corrections, which introduce infra-red (IR) divergencies, Fig. 1.3b.
- We can radiate additional particles, which introduce soft and/or collinear divergencies, e.g. Fig. 1.3c.
- The initial parton can also be a gluon now, e.g. Fig. 1.3d.

As requested in conclusion 1.2 we have to sum over all possible diagrams and since we only consider the fully inclusive case here, we can apply the Kinoshita-Lee-Nauenberg theorem [27–29] straightforwardly and cancel all soft and IR divergencies. Finally, we apply collinear factorization and subtract the remaining collinear divergencies from the coefficient functions and reabsorb them into the PDF definition, Eq. (1.1). Any divergency which we remove introduces a distribution, such as the Dirac delta distribution or the usual $+$-distribution, into the coefficient functions, i.e. there is a direct mapping.

Eventually, we get for all quarks q

$$\begin{aligned} C_q^{(1)}(z) &= e_q^2 C_F \left[4\left(\frac{\ln(1-z)}{1-z}\right)_+ - 3\left(\frac{1}{1-z}\right)_+ - (9+4\zeta_2)\delta(1-z) \right. \\ &\qquad \left. -2(1+z)\ln\left(\frac{1-z}{z}\right) - 4\frac{\ln(z)}{1-z} + 6 + 4z \right] \qquad (1.17) \\ &= C_{\bar{q}}^{(1)}(z) \qquad (1.18) \end{aligned}$$

and for the gluon

$$C_g^{(1)}(z) = \left(\sum_q e_q^2\right) \left[(2-4z(1-z))\ln\left(\frac{1-z}{z}\right) - 2 + 16z(1-z) \right] \qquad (1.19)$$

where C_F is the second Casimir operator of the fundamental representation, $C_F = (N_C^2-1)/(2N_C)$, with $N_C = 3$ the number of colors.

At a given perturbative order we explore a certain space of mathematical (special) functions and, e.g., at NLO we encounter for the first time the Riemann Zeta function $\zeta(n)$, which is defined at integer values n by

$$\zeta_n = \zeta(n) = \sum_{j=1}^{\infty} \frac{1}{j^n} \,. \tag{1.20}$$

The definition of the usual $+$-distribution is given in Eq. (2.6) and discussed further in Section 2.2.1.

Chapter 2.

DIS in real life

In the following we expose the complexity that DIS requires, when we want to confront actual experimental measurements with our theoretical predictions, or the other way round, if we want to use them to extract realistic PDFs. We start by reviewing a small part of the available data in Section 2.1, which introduces new particles and thus new possibilities into the game. In Section 2.2 we examine how to implement the coefficient function in an efficient way, such that PDF fitting groups can use them in a simple way. We review the pQCD property of resummation in Section 2.3, which we then use two discuss two other major features: first, scale variations in Section 2.4 and then the treatment of heavy quarks in Section 2.5. Kinematic corrections are discussed in Section 2.6 and in Section 2.7 we turn instead to more theoretical considerations. Finally, in Section 2.8 we collect all considered corrections and spell the hidden variables in the textbook version of Eq. (1.1).

All topics are either explicitly or at the very least always implicitly correlated with each other, making a given feature actual more complicated then it may seem on first glance. Thus the sorting of the sections is challenging, but hopefully follows now an order of importance. When appropriate we try to spell the interplay explicitly.

2.1. DIS is diverse

In this section we give an overview of some of the different DIS datasets commonly used in PDF fits. The various experiments probe different physics and are thus necessary for a realistic PDF extraction.

Figure 2.1.: "The view into the 6.3-kilometre-long HERA tunnel shows the superconducting magnets for guiding the protons on top and the electron ring beneath them." - taken from [30]

2.1.1. HERA

We start with the Hadron-Elektron-Ringanlage (HERA), which was in operation between 1992 and 2007. HERA was hosted by DESY in Hamburg, Germany and Fig. 2.1 shows an image from the HERA tunnel. While the promoted physics discovery, leptoquarks, eventually did not happen, HERA has become one of the most influential DIS machines so far. In fact, it is until this day the only lepton-proton collider in the world. It recorded a huge amount of data, which is still analyzed nowadays, 20 years after the shutdown. HERA had four experiments (H1, ZEUS, HERA-B and HERMES), but we focus here on the combined analysis of H1 and ZEUS [31]. In Table 2.1 we given an overview of the most important fully-inclusive DIS measurements from HERA.

We start by looking at the projectile column and immediately we realize that HERA was most of the time not colliding electrons, but positrons. This is to remind us that DIS is lepton-hadron scattering and, indeed, we have to consider all leptons. We expand our selection further below in Section 2.1.2, but remain for the moment with the electron and the positron, which are exact copies of each other, but with opposite charge.

Next, we turn to the energy column $\sqrt{s}$, which spreads across four different energies. In practice the lepton beam was operated with a fixed beam energy ($E_e = 27.5\,\text{GeV}$) and varying energies for the protons ($E_p = 920\,\text{GeV},\ 820\,\text{GeV},\ 575\,\text{GeV}$ and $460\,\text{GeV}$).

Table 2.1.: Fully inclusive lepton-proton scattering measurements at HERA [31]. We quote the scattered lepton, the measured kinematic ranges, the total number of data points N_{dat}, and the measured observable. $N_{\text{dat}}^{\text{fit}}$ refers to the number of data points used in NNPDF4.0 [9].

projectile	$\sqrt{s}$ [GeV]	x	Q^2 [GeV^2]	$N_{\text{dat}}^{\text{fit}}/N_{\text{dat}}$	obs.
e^-	318	0.0008–0.65	60–5×10^4	$159/159$	σ_r^{NC}
e^+	318	5.02×10^{-6}–0.65	0.15–3×10^4	$377/485$	σ_r^{NC}
e^+	300	6.21×10^{-7}–0.4	0.045–3×10^4	$70/112$	σ_r^{NC}
e^+	251	2.79×10^{-5}–0.65	1.5–800	$254/260$	σ_r^{NC}
e^+	225	4.64×10^{-5}–0.65	2–800	$204/209$	σ_r^{NC}
e^-	318	0.008–0.65	3×10^2–3×10^4	$42/42$	σ_r^{CC}
e^+	318	0.008–0.4	3×10^2–3×10^4	$39/39$	σ_r^{CC}

If we then look to the x and Q^2 columns and recall Eq. (1.7), we see that the table only indicates the maximal available range of those variables. Specifically, HERA was able to measure activity in the detector for $0.005 < y < 0.95$.

However, not only the accessible kinematical phase space limits our kinematic coverage, but also our requirement to study DIS, conclusion 1.6. In practice, e.g. NNPDF4.0 [9] applies a cut of $Q^2 \geq 8\,\text{GeV}^2$ and $W^2 \geq 12.5\,\text{GeV}^2$. Mathematically, the Q^2 cut ensures we are in a perturbative regime, Eq. (1.15), while the W^2 cut has multiple motivations. First, it ensures we are in an inelastic regime, i.e. we are not finding resonances or excitations of the proton. Second and related to the former, we know it suppresses the correction term $\mathcal{O}\left(\frac{\Lambda_{\text{QCD}}^2}{Q^2}\right)$ in Eq. (1.1). Studying and trying to estimate the correction term is a topic in current research [32–34] and we return to it in Section 2.6. Looking to the column with the number of data points, we note that even after the kinematic cuts we are left with a total of 1145 measured fully inclusive DIS cross sections.

Finally, we turn to the observables column for which we see two different entries exist: the reduced neutral current (NC) cross section σ_r^{NC} and the reduced charge current (CC) cross section σ_r^{CC}. The "current" here refers to the current between the leptonic part and the hadronic part, in particular the exchanged boson. The "reduced" cross section refers to a slight rescaling of the double differential cross section, Eq. (1.13),

$$\sigma_r = \frac{xyQ^2}{4\pi\alpha_{\text{em}}^2}\frac{\mathrm{d}^2\sigma}{\mathrm{d}x\mathrm{d}y} \tag{2.1}$$

such that it refers to a specific PDF combination at LO accuracy.

Next, we examine the CC case by recalling two important facts from conclusion 1.1: first, charged leptons, and thus also electrons and positrons, have weak charges and, second, W bosons have an electric charge. Thus, by CC we consider the lepton interacts with the hadron via the exchange of a W boson, which behaves quite different then a photon. First and foremost, due to the electric charge conservation the W boson can only interact with specific leptons and quarks. On the leptonic side this implies that, first, e.g., the positron always emits a W^+ boson, and, second, the recoiling lepton is a neutrino, which in practice is impossible to track. Specifically, it is impossible to measure it's momentum four-vector k', Eq. (1.5), and thus measure the W boson momentum q, Eq. (1.6), or any other derived variable (x, Q^2, y). Instead, the full reconstruction of the hadronic final state also allows to extract the relevant kinematic variables, which is used instead [31]. Although the production of CC events is suppressed by the propagator of the W boson, $\sim (Q^2 + m_W^2)^{-1}$, it is a LO effect and thus no further suppression. Moreover, since the W are maximally parity violating this allows a third, independent structure function (typically referred to as F_3) to appear in the decomposition of the cross section, Eq. (1.13). We review the problems introduced by γ_5, which causes the parity violation, further in Section 3.1.

We now consider the NC case, which is a generalization of the photon exchange discussed in Chapter 1. As we recall from conclusion 1.1 the Z boson is practically a massive copy of the photon γ and they share all other quantum numbers. This implies whenever a photon is exchanged, also a Z boson can be exchanged, with the main difference that its contributions are suppressed by the respective propagator, $\sim (Q^2 + m_Z^2)^{-1}$. More specifically, we get three contributions to our final cross section: a pure photon contribution σ^γ (where the superscript γ is often suppressed), a pure Z contribution σ^Z, and an interference contribution[1] $\sigma^{\gamma Z}$, which have to be summed together:

$$\sigma^{\rm NC} = \sigma^\gamma + \sigma^{\gamma Z} + \sigma^Z\,. \tag{2.2}$$

While at low virtualities $Q^2 \ll m_Z^2$ the Z contributions can be neglected, this does not apply to the largest HERA measurements. Similar to W boson, also Z bosons have an axial-vectorial coupling to fermions, $\sim \gamma^\mu\gamma^5$, which introduces a contribution from F_3 to Eq. (1.13). The interference contribution, $F_{1,2}^{\gamma Z}$, yields an electric charge (from the photon) and a vectorial coupling (from the Z boson) both on the leptonic and the hadronic side, whereas the pure Z contribution, $F_{1,2}^Z$, yields squares of both vectorial and

[1] the interference term typically contains the required factor of 2 from the binomial formula inside

axial-vectorial couplings. On the other side, as F_3 is per-se a parity-violating observable, there the interference contribution $F_3^{\gamma Z}$ yields an electric charge (from the photon) and an axial-vectorial coupling (from the Z) and the pure Z contribution, F_3^Z yields a product of the vectorial and axial-vectorial couplings.

The fact that NC and CC couple completely different to fermions, i.e. quarks and leptons, and the definition of three independent structure functions in both cases allows one to fully determine PDFs from DIS [35]. We discuss more technical details in Section A.1.

Let us return to the key point of this section here. Based on the discussion above, we can now better justify why we don't consider all particles in Fig. 1.1. According on our definition of DIS, conclusion 1.4, in theory also the Higgs boson would contribute to NC cross section, but since its coupling is proportional to the mass of the particle and we always[2] consider massless leptons it does in practice not contribute. As the tau has a very short lifetime they are in practice impossible to accelerate in a collider and thus not suitable for DIS collider experiments. Instead, while muons are also difficult to accelerate concrete experiments for muon-proton scattering have been proposed [37].

2.1.2. Nuclear and Neutrino DIS

After having considered all available bosons and a few more leptons, we now need to consider more hadrons. While we focus here on the proton PDF point-of-view, DIS is equally important for nuclear PDFs [38–41], which describe the internal structure of nuclei. Moreover, the two are typically directly intertwined: proton PDF extractions often contain measurements on nuclear target [9], while nuclear PDF extraction often require a proton baseline.

- CHORUS, Sami
- FASER
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[2] almost always [36] - but even so Higgs would only contribute a very large Q^2

2.2. Code implementation

If we want to compare any theoretical prediction to an experimental measurement we eventually need to implement the coefficient functions into a computer program. Specifically, we need to find a representation of our results, which is suitable for computers and which can be evaluated numerically, preferably in a fast and efficient way.

2.2.1. Distributions

Coefficient functions may contain distributions, for example the quark coefficient function at LO is just a Dirac delta distribution:

$$C_q^{(0)}(z) = e_q^2 \delta(1-z)\,. \tag{2.3}$$

To better compare with our next step, let's repeat the definition of the Dirac delta distribution:

$$\int_0^1 \mathrm{d}z\, \delta(1-z) f(z) = f(1). \tag{2.4}$$

However, this is only the simplest case, which appears at LO, and at higher order we get more complicated distributions. For example the quark coefficient function at NLO, Eq. (1.17), contains many more distributions

$$C_q^{(1)}(z) \supset 4\left(\frac{\ln(1-z)}{1-z}\right)_+ - 3\left(\frac{1}{1-z}\right)_+ - (9+4\zeta_2)\delta(1-z) \tag{2.5}$$

where the usual +-distributions are defined via

$$\int_0^1 \mathrm{d}z\, a(z)_+ f(z) = \int_0^1 \mathrm{d}z\, a(z)\left(f(z) - f(1)\right)\,. \tag{2.6}$$

We review in Section 1.3 briefly how these distributions appear, but here we turn to more practical matters. The problem is: distributions are (complicated) math objects, defined only via integral relations, Eqs. (2.4) and (2.6), which are unsuitable for numeric evaluation. Moreover, Eq. (2.6) is defined via an integral spanning the whole domain of

z, i.e. in particular, the lower integration bound is 0, but Eq. (1.1) requires a convolution, where instead the lower integration bound depends on the measured kinematics x.

Thus, it is convenient to introduce a different representation for coefficient functions, the Regular-Singular-Local (RSL) representation. In the RSL representation any coefficient function $C(z)$ is mapped onto three functions,

$$C(z) \leftrightarrow (C^R(z), C^S(z), C^L(x)) \tag{2.7}$$

where $C(z)$ may contain distributions, but neither of $C^R(z)$, $C^S(z)$, or $C^L(x)$. The three functions are defined via their action under the convolution integral,

$$(C \otimes f)(x) = \int_x^1 \frac{dz}{z} f(x/z) C^R(z) + \int_x^1 dz \left(\frac{f(x/z)}{z} - f(x)\right) C^S(z) + f(x) C^L(x) \tag{2.8}$$

for any sufficiently smooth test function f. Note that Eq. (2.8) only contains integrals with the lower integration bound x and that all terms are finite if all RSL functions are finite. Also note that while the argument of C^S and C^L is intentionally called z, referring to the partonic momentum fraction over which we integrate, the argument of C^L is denoted by x, which refers to the argument of the convolution on the right-hand-side.

Let us consider some simple examples.

- If the coefficient function is a regular function $C(z) = r(z)$ we find

$$C^R(z) = r(z), \qquad C^S(z) = 0, \qquad C^L(x) = 0 \tag{2.9}$$

- If the coefficient function is a Dirac delta distribution $C(z) = \delta(1-z)$ we find

$$C^R(z) = 0, \qquad C^S(z) = 0, \qquad C^L(x) = 1 \tag{2.10}$$

- If the coefficient function is a "raw" +-distribution $C(z) = (p(z))_+$ we find

$$C^R(z) = 0, \qquad C^S(z) = p(z), \qquad C^L(x) = -\int_0^x dz\, p(z) \tag{2.11}$$

More examples can be found in Section A.2. Eventually, we also face the problem that +-distributions can absorb regular functions. For example Eq. (1.17) can also be written as

$$C_q^{(1)}(z) = e_q^2 C_F \left[4\frac{\ln(1-z)}{1-z} - 3\frac{1}{1-z} - 2(1+z)\ln\left(\frac{1-z}{z}\right) - 4\frac{\ln(z)}{1-z} + 6 + 4z\right]_+ \tag{2.12}$$

where we have absorbed all functional dependence under the distribution. Thus before changing the representation we also need to fix the actual form and one possible choice is to only allow the minimal functional dependence under the distribution sign, e.g. as in Eq. (1.17).

2.2.2. Interpolation

The RSL representation together with a suitable implementation of Eq. (2.8) provides already a practical way to implement arbitrary convolutions. However, we can make an additional step by applying a further discretization, which is desirable for computers. In the following we only consider the dependence on either the Bjorken-x or its the momentum fraction counterparts, z or ξ, and drop any scale dependence Q^2 for the remainder of the section.

We assume that there is a minimum momentum fraction $x_{\text{grid,min}}$ for which we need to compute observables, which implies that we also only need coefficient functions $\mathbf{C}(z)$ and PDFs $\mathbf{f}(z)$ above this limit, Eq. (1.1). Next, we choose a set of N_{grid} grid points,

$$\mathbb{G} = \{x_k : x_{\text{grid,min}} < x_k <= 1, \quad k = 0, \dots, N_{\text{grid}}\} \tag{2.13}$$

which we then use to interpolate the PDF

$$f_j(\xi) \sim \sum_{k=0}^{N_{grid}-1} f_j(x_k) p_k(\xi) \tag{2.14}$$

with a suitable set of interpolation polynomials $p_k(\xi)$. In practice, piecewise Lagrange interpolation polynomials are a good choice [42, 43].

Inserting Eq. (2.14) into Eq. (1.1) yields

$$\sigma(x) = \sum_j \int_x^1 \frac{\mathrm{d}z}{z} C_j(z) \left(\sum_{k=0}^{N_{grid}-1} f_j(x_k) p_k(x/z) \right) \tag{2.15}$$

$$= \sum_j \sum_{k=0}^{N_{grid}-1} f_j(x_k) \hat{\sigma}_{jk}(x) \tag{2.16}$$

where we defined

$$\hat{\sigma}_{jk}(x) = \big(C_j \otimes p_k\big)(x)\,. \tag{2.17}$$

Choosing our grid $\mathbb{G}$ fixes immediately the interpolation polynomials and thus we can tabulate $\hat{\sigma}_{jk}(x)$ without having knowledge about the actual PDF $f_j(\xi)$. Note that the interpolation in momentum fraction space is independent from the flavor index j and for simplicity we can choose the same grid for all flavors. This makes both $f_j(x_k)$ and $\hat{\sigma}_{jk}(x)$ two two-dimensional tensors in Eq. (2.16), which get multiplied in such a way that they yield a plain number.

2.2.3. Grids

Being able to compute $\hat{\sigma}_{jk}(x)$ without knowledge about the actual PDF is advantages since, we recall, PDFs are non-perturbative objects, encoding low-scale physics, which are not computable from first principles inside pQCD. Instead, PDFs are obtained from a fitting procedure, which can be summarized as follows:

Conclusion 2.1: PDF fitting

1. Guess a candidate PDF
2. Compute the theory predictions with this candidate PDF
3. Compare the theory predictions to the experimental measurement

Repeat the above steps until the procedure converges, i.e. the comparison is "good enough".

The specific details on how to guess a candidate PDF, how to exactly compute the theory predictions, which experimental data to use, and what "good enough" exactly means is a choice. This is the reason why competing PDF fitting groups exist and why they do not necessarily obtain the same PDF, although we all assume the latter to be universal.

The main point for us here is that we need to compute theory predictions very many times for slightly different candidate PDFs. This is a serious problem for complicated hadronic observables, such as, e.g., jet cross sections [44], where the computation can take many hours. The solution is to compute fast interpolation grids, using, e.g., PineAPPL [43, 45], APPLgrid [46], or fastNLO [47], which store the partonic matrix elements in an efficient way, such that any subsequent convolution with an arbitrary PDF is fast. Having all observables in a PDF fit sharing the same technology simplifies their consistent treatment and inside the NNPDF collaboration this is realized in the pineline framework [48]. In practice we organize discretization of the coefficient functions in a slightly different way.

Specifically, interpolation grids are higher dimensional tensors which we can recombine in a suitable way a posteriori by performing fast linear algebra operations. The first three major dimensions are:

bin This is the list of cross section that we need to compute simultaneously. In DIS this typically corresponds to all N_{dat} data point measured in a given configuration, e.g. in Table 2.1. Thus, in practice the actual list is given by the experimental measurement.

order This is the list of perturbative orders, i.e. we store each $\mathbf{C}^{(k)}(z)$, Eq. (1.15), separately. Having the full list available allows to use a given grid also at lower orders and thus study, e.g., perturbative stability. In practice, the actual list is given by the requested accuracy of the user (provided we can actual fulfill that request).

channel This is the list of PDF flavor combinations, i.e. the sum over j in Eq. (1.1). Having the full list available allows to study the impact of different PDF combination and, e.g., determine which PDF the observable can potentially constrain most. In DIS it is given by the coupling structure of the exchanged boson, i.e. γ/Z vs. $W^{\pm}$. In practice, the actual list is the easiest given directly in terms of individual parton flavors as the available combinations are too complex (see Section 2.4 for additional complications).

These three dimensions are present in any interpolation grid and in DIS there are two additional dimensions:

scale This is the list of factorization and renormalization scales, discussed in Section 2.4. In DIS this dimension turns out to be trivial, i.e. it has support by a single value, as in fully inclusive DIS there is only a single scale, Q^2 (or a multiplicative there of if we do not set $\mu_F^2 = \mu_R^2 = Q^2$). In practice, the actual list is given by the Q^2 specified in the bin configuration.

momentum fraction This is the list of momentum fractions, Eq. (2.13), over which the coefficient functions are convolved with the PDF. Recall that LO, Eq. (1.16), and NLO, e.g. Eq. (1.17), are quite different and thus they use different ranges of momentum fractions. In practice, the actual list is given by a convolution with a suitable set of interpolation functions, which form a basis for PDFs.

For fully inclusive DIS we end up with a five dimensional tensors, where each inner dimension may depend on the outer.

2.3. Resummation

In preparation to Sections 2.4 and 2.5 we need to review two main ingredients into pQCD: the strong coupling $\alpha_s(Q^2)$, Eq. (1.15), and the PDFs $\mathbf{f}(\xi, Q^2)$, Eq. (1.1). Specifically, we need to remind ourselves on their scale dependency also referred to as evolution. While the two objects obey very similar equations, the strong coupling $\alpha_s(Q^2)$ is conceptually simpler and we start with that.

2.3.1. The strong coupling

The strong coupling α_s is directly linked to the coupling of the gluons to quarks,

$$\mathcal{L}_{\text{QCD}} \supset gT_a\bar{\psi}\gamma^\mu A_\mu^a\psi, \qquad \alpha_s = \frac{g^2}{4\pi} \tag{2.18}$$

and is the central object in pQCD around which we organize our perturbative series. If one considers higher order corrections to the gluon-quark couplings, one encounters quickly divergencies of ultra-violet origin, i.e. for large loop momenta. This forces the introduction of a regularization procedure, typically dimensional regularization, which eventually leads to a scale dependent definition of the renormalized strong coupling $\alpha_s(\mu_R^2)$. The introduced scale, called renormalization scale μ_R henceforth, is an arbitrary scale, which is an artifact of our regularization procedure and which effectively represents a separation between perturbative and non-perturbative regimes. However, the scale dependence of $\alpha_s(\mu_R^2)$ is rigorously predicted in pQCD, organizing the terms in such a way that within the requested accuracy any dependence of μ_R is cancelled. Recall that the scale dependence can be computed once the Lagrangian $\mathcal{L}$ is defined and does not require any actual observable.

Eventually, we obtain the renormalization group equation (RGE) of the strong coupling,

$$\mu_R^2 \frac{\mathrm{d}\alpha_s(\mu_R^2)}{\mathrm{d}\mu_R^2} = \beta(\alpha_s) = -\sum_{j=0} \beta_j \big(\alpha_s(\mu_R^2)\big)^{2+j} \tag{2.19}$$

which is also the definition of the beta function $\beta(\alpha_s)$. The upper limit of the sum in Eq. (2.19) defines the perturbative accuracy, to which all other objects in pQCD refer to.

At LO accuracy we find

$$\mu_R^2 \frac{\mathrm{d}\alpha_s(\mu_R^2)}{\mathrm{d}\mu_R^2} = -\beta_0 \big(\alpha_s(\mu_R^2)\big)^2, \qquad \beta_0 = \frac{11C_A - 4T_R n_f}{12\pi} \tag{2.20}$$

where we used the second Casimir constant of the adjoint representation $C_A = N_c = 3$, the normalization of the fundamental generators $T_R = 1/2$, and the number of active flavors n_f, discussed in Section 2.5. In this specific case we can solve Eq. (2.20) and obtain

$$\alpha_s^{\mathrm{LO}}(\mu_R^2) = \frac{\alpha_s(\mu_{R,0}^2)}{1 + \alpha_s(\mu_{R,0}^2)\beta_0 \ln\big(\mu_R^2/\mu_{R,0}^2\big)} \tag{2.21}$$

for a given boundary condition $\alpha_s(\mu_{R,0}^2)$ at the scale $\mu_{R,0}^2$. Recall, that Eq. (2.21) has not two free parameters, $\mu_{R,0}$ and $\alpha_s(\mu_{R,0})$ as one would naively think, but only one Λ_{QCD} (see exercise A.3.1). Also remember that Λ_{QCD} corresponds to the pole in Eq. (2.21) and pQCD only makes sense above that scale, $\mu_R > \Lambda_{\mathrm{QCD}}$. Moreover, in practice it is often convenient to rewrite Eq. (2.19) in terms of $t_R = \ln\big(\mu_R^2/\mu_{R,0}^2\big)$.

Now, coming back to the title of this section, note that Eq. (2.21) is a resummation:

$$\alpha_s^{\mathrm{LO}}(\mu_R^2) = \alpha_s(\mu_{R,0}^2) \sum_{k=0}^{\infty} \big(-\alpha_s(\mu_{R,0}^2)\beta_0 \ln\big(\mu_R^2/\mu_{R,0}^2\big)\big)^k \tag{2.22}$$

i.e. it collects an infinite tower of logarithms, here $\ln\left(\mu_{R,0}^2/\mu_R^2\right)$, into a compact expression, Eq. (2.21). Since $\mu_{R,0}$ is an arbitrary scale, often chosen to coincide with the mass of the Z boson $\mu_{R,0} = m_Z$, it is basically always required to perform the resummation and use Eq. (2.21). However, in some situations, to be discussed in Section 2.4, we need Eq. (2.21) truncated to a finite order, i.e. unraveling Eq. (2.22), and we get

$$\alpha_s^{\mathrm{LO}}(\mu_R^2) = \alpha_s(\mu_{R,0}^2) - \left(\alpha_s(\mu_{R,0}^2)\right)^2 \beta_0 \ln\left(\mu_R^2/\mu_{R,0}^2\right) + \ldots \tag{2.23}$$

While for the special case $\mu_{R,0} = \mu_R$ we need to be consistent of course and we thus find that $\alpha_s^{\mathrm{LO}}(\mu_R^2)$ is $\alpha_s(\mu_{R,0}^2)$ at the LO truncation, we note that they differ by higher order terms. Equation (2.23) is the first, simplest case of a perturbative expansion of a resummed expression, Eq. (2.21), and can also be obtained by using the Taylor expansion of $\alpha_s(\mu_R^2)$ around $\mu_{R,0}^2$ together with Eq. (2.20). In other words, Eq. (2.23) is a perturbative expansion of Eq. (2.21) in powers of $\alpha_s(\mu_{R,0}^2)$. Note that Eq. (2.22) has a finite radius of convergence (as all geometric series) and that is also reflected in Eq. (2.21) by the pole Λ_{QCD}.

Starting from NLO accuracy the RGE, e.g. in this specific case given by

$$\mu_R^2 \frac{\mathrm{d}\alpha_s(\mu_R^2)}{\mathrm{d}\mu_R^2} = -\beta_0\left(\alpha_s(\mu_R^2)\right)^2 - \beta_1\left(\alpha_s(\mu_R^2)\right)^3 , \tag{2.24}$$

can no longer be solved exactly with elementary functions. However, it is still possible to find an analytic solution which solves Eq. (2.24) at the requested accuracy, i.e. in this case NLO. FH: exercise: proof NLO alphas? This solution still performs a resummation, FH! as we have to, but it only collects the so-called leading logarithms. FH: show exp alphas FH! is LL A completely different way to solve Eq. (2.24) is using numerical tools instead. We refer to the literature on how to implement a specific solution in practice and remain here just with the fact that it is a complicated problem on its own, which has no unique solution and for which approximations are mandatory.

Conclusion 2.2: Solution of RGE

- The LO RGE of the strong coupling, Eq. (2.20), can be solved exactly, Eq. (2.21)
- The RGE of the strong coupling at higher orders, e.g. Eq. (2.24), can not be solved exactly and one needs to choose a solution method

Any proper solution of the RGE must solve the RGE itself at the requested accuracy else it is not a solution, but they can and do differ by terms beyond - we refer to the uncertainty which is associated with this freedom as Missing Higher Order Uncertainty (MHOU). Truncating the perturbative series is a fundamental part of pQCD and thus we are always bound to the accuracy which we request at the beginning. While in many simple application MHOU are often neglected they are an important ingredient in modern PDF fits [49] and in Section 2.4 we demonstrate one possible way to estimate them.

2.3.2. DGLAP

For PDFs $\mathbf{f}(\xi, \mu_F^2)$ the situation is essentially the same as for the strong coupling $\alpha_s(\mu_R^2)$, except the latter is just a number and the former are a set of functions. The scale dependence is a again predicted by pQCD, but now we need an arbitrary physical observable to do so. The specific observable does not matter since PDFs are universal, but due to the linear dependence on PDF and the "simplicity" of fully inclusive DIS it often serves as reference case. The arbitrary scale that is introduced in the regularization procedure, typically called collinear factorization, is referred to as factorization scale μ_F henceforth.

Eventually, we obtain the RGE of PDFs, often referred to as DGLAP equation named after the researcher who established it [50–52],

$$\mu_F^2 \frac{\mathrm{d}\mathbf{f}(\xi, \mu_F^2)}{\mathrm{d}\mu_F^2} = \left(\mathbf{P}(\mu_F^2) \otimes \mathbf{f}(\mu_F^2)\right)(\xi) = \sum_{j=0} \left(\alpha_s(\mu_F^2)\right)^{1+j} \left(\mathbf{P}^{(j)} \otimes \mathbf{f}(\mu_F^2)\right)(\xi) \tag{2.25}$$

where $\mathbf{P}(y, \mu_F^2)$ is the matrix of Altarelli-Parisi splitting functions. Note that the upper limit of the sum in Eq. (2.25) must be set to the requested perturbative accuracy, i.e. consistent with Eq. (2.20).

When attempting to solve Eq. (2.25) we must worry about the two complications in PDFs with respect of the strong coupling. First, since PDFs are a vector and thus the Altarelli-Parisi splitting functions are combined in a matrix $\mathbf{P}$, we need to worry about non-commutativity. Indeed, while at LO accuracy we only have a single matrix $\mathbf{P}^{(0)}$, which trivially commutates with itself, starting from NLO we encounter a second matrix $\mathbf{P}^{(1)}$, which does not commutate with the former. Second, at any fixed scale PDFs are a function of the momentum fraction ξ and, moreover, the right-hand-side of Eq. (2.25) features an integral (with the definition of the convolution $\otimes$ in Eqs. (1.1) and (1.2)).

To overcome the second problem, one possibility is to apply an integral transformation, the so-called Mellin transformation, defined by

$$\tilde{\mathbf{f}}(N, \mu_F^2) = \int_0^1 \mathrm{d}\xi\, \xi^{N-1} \mathbf{f}(\xi, \mu_F^2) \tag{2.26}$$

$$\gamma(N, \mu_F^2) = \int_0^1 \mathrm{d}y\, y^{N-1} \mathbf{P}(y, \mu_F^2) \tag{2.27}$$

where the Mellin-transformed splitting functions γ are often referred to as anomalous dimensions. Inserting Eqs. (2.26) and (2.27) into Eq. (2.25) yields a "simple" first order differential equation, which can be solved by a "simple" exponentiation. However, in practice PDFs are mostly needed in momentum fraction space, $\mathbf{f}(\xi, \mu_F^2)$, and not in Mellin space, $\tilde{\mathbf{f}}(N, \mu_F^2)$, thus if we use the Mellin transform, we must also use its inverse transformation.

We can write the general solution to Eq. (2.25) in terms of an Evolution Kernel Operator (EKO) [42] $\mathbf{E}$,

$$\mathbf{f}(\xi, \mu_F^2) = \left(\mathbf{E}(\mu_F^2 \leftarrow \mu_{F,0}^2) \otimes \mathbf{f}(\mu_{F,0}^2)\right)(\xi)\,, \tag{2.28}$$

but we skip spelling out an explicit solution here and refer to the literature instead [42]. The important part for us is that the operator (which is almost an exponential) again captures a resummation, in this case of $\ln\left(\mu_F^2/\mu_{F,0}^2\right)$. Again, we can perform a perturbative expansion of the LO resummed solution to see the logarithm explicitly,

$$\mathbf{f}^{\mathrm{LO}}(\xi, \mu_F^2) = \mathbf{f}(\xi, \mu_{F,0}^2) + \alpha_s(\mu_{F,0}^2) \ln\left(\mu_F^2/\mu_{F,0}^2\right) \left(\mathbf{P}^{(0)} \otimes \mathbf{f}(\mu_{F,0}^2)\right)(\xi) + \ldots \tag{2.29}$$

and starting from NLO accuracy, we face the same issues as for the strong coupling and conclusion 2.2 applies in the same way.

2.4. Scale variations

We now attempt to address a problem we first see in Section 2.3: Missing Higher Order Uncertainty (MHOU). We repeat again, the truncation of the perturbative series is a fundamental part of our framework and we are bound to it. The naive solution of just computing the next higher correction is not a solution, as we would face the problem just again there. Thus the task is to get an estimate of the missing terms based on current knowledge, which by definition is an ill-posed problem. Several possible strategies have been suggested in the literature [49, 53–59], which are based on resummation of specific variables and/or statistical arguments.

From the practical point of view as a PDF fitter, however, there is one big problem with many of the proposed algorithms: they only work for some specific set of observables and require a detailed analysis of the kinematics. Thus, we review here the most common approach to estimate MHOU, which can be rigorously derived in pQCD and applied to any observable: scale variations. On the other side, especially in recent years, scale variations have been challenged as suitable tool, due to the inherent arbitrariness and the lack of statistical interpretation (to which we return in the end). For now we focus on their practical implementation in DIS.

2.4.1. Renormalization scale

We start again from the simpler case of the renormalization scale μ_R associated with the strong coupling $\alpha_s(\mu_R^2)$ via the RGE, Eq. (2.20). As we say above, the renormalization scale captures the ultra-violet divergencies, which arise, e.g., from self-energy diagrams and which are thus associated with the partonic diagram itself. Moreover, as we see from Eq. (1.15), we expand the partonic coefficient functions in powers of the strong coupling.

Conclusion 2.3: Renormalization scale dependence

The renormalization scale dependence is captured by the partonic coefficient function $\mathbf{C}(z, Q^2, \mu_R^2)$.

Since we have to choose something for the arbitrary scale μ_R and we only have a single scale available, the virtuality Q^2, we start by setting them equal to each other $\mu_R^2 = Q^2$, which is also the implicit choice in Eq. (1.1). We then need to find a way to estimate the missing terms in $\mathbf{C}(z, Q^2, \mu_R^2 = Q^2)$, i.e. something that is identical at the requested perturbative accuracy, but differs from it beyond. This can be achieved by using Eq. (2.23) together with a suitable redefinition of the perturbative coefficients of the partonic coefficient function $\mathbf{C}^{(j)}(z)$. Specifically, we require

$$\mathbf{C}(z, Q^2, \mu_R^2) = \mathbf{C}(z, Q^2, \mu_R^2 = Q^2)\big(1 + \mathcal{O}\big(\alpha_s(Q^2)\big)\big) \tag{2.30}$$

with

$$\mathbf{C}(z, Q^2, \mu_R^2) = \sum_{j=0} \big(\alpha_s(\mu_R^2)\big)^j \mathbf{C}^{(j)}(z, \ln\big(\mu_R^2/Q^2\big)) \,. \tag{2.31}$$

Before we continue with deriving explicit expressions for $\mathbf{C}^{(j)}(z, \ln\big(\mu_R^2/Q^2\big))$ some comments are in order:

- $\mathbf{C}(z, Q^2)$ in Eq. (1.1) is an abbreviation of $\mathbf{C}(z, Q^2, \mu_R^2 = Q^2)$ and $\mathbf{C}^{(j)}(z)$ of $\mathbf{C}^{(j)}(z, 0)$
- in Eq. (2.31) μ_R has its intended meaning: it is the scale of the strong coupling
- from Section 2.3.1 we know $\alpha_s(\mu_R^2)$ is a pure resummation of logarithms, thus $\mathbf{C}^{(j)}$ may only depend on $\ln\big(\mu_R^2/Q^2\big)$
- $\mathbf{C}^{(j)}(z, 0)$ can be computed from diagrams by assuming explicitly $\mu_R = Q$ there

We now derive the explicit next-to-next-to-leading order (NNLO) expression for fully inclusive DIS. In particular, we recall that LO is given by $\mathbf{C}^{(0)}(z)$, i.e. proportional to $\big(\alpha_s(Q^2)\big)^0$ in Eq. (1.15), NLO is given by $\mathbf{C}^{(1)}(z)$ proportional to $\big(\alpha_s(Q^2)\big)^1$, and NNLO by $\mathbf{C}^{(1)}(z)$ proportional to $\big(\alpha_s(Q^2)\big)^2$. Moreover, we assume that the coefficient functions at the central scale $\mu_R = Q$ are known and we can write Eq. (1.15) up to NNLO

$$\mathbf{C}(z, Q^2) = \mathbf{C}^{(0)}(z) + \alpha_s(Q^2)\mathbf{C}^{(1)}(z) + \big(\alpha_s(Q^2)\big)^2 \mathbf{C}^{(2)}(z) \tag{2.32}$$

Expanding Eq. (2.31) up to the same accuracy we get

$$\begin{aligned} \mathbf{C}(z, Q^2, \mu_R^2) = {} & \mathbf{C}^{(0)}(z, \ln\big(\mu_R^2/Q^2\big)) + \alpha_s(\mu_R^2)\mathbf{C}^{(1)}(z, \ln\big(\mu_R^2/Q^2\big)) \\ & + \big(\alpha_s(\mu_R^2)\big)^2 \mathbf{C}^{(2)}(z, \ln\big(\mu_R^2/Q^2\big)) \end{aligned} \tag{2.33}$$

We are now left with the task to make Eq. (2.32) and Eq. (2.33) perturbatively equivalent, and although we are considering NNLO the equivalence must hold at any lower order. Thus, we first can consider LO: since there is no dependency on the strong coupling in the first place, the scale-varied coefficient function may not depend on the renormalization scale and we find

$$\mathbf{C}^{(0)}(z, \ln\big(\mu_R^2/Q^2\big)) = \mathbf{C}^{(0)}(z) \,. \tag{2.34}$$

Starting from NLO an explicit dependency on the strong coupling appears, however, from Eq. (2.23) we know that the two choices, $\alpha_s(Q^2)$ and $\alpha_s(\mu_R^2)$ differ by higher order terms. Specifically, we have

$$\alpha_s(Q^2) = \alpha_s(\mu_R^2) + \mathcal{O}\Big(\big(\alpha_s(\mu_R^2)\big)^2\Big) \quad \text{and} \quad \alpha_s(\mu_R^2) = \alpha_s(Q^2) + \mathcal{O}\Big(\big(\alpha_s(Q^2)\big)^2\Big) \tag{2.35}$$

Thus, if Eq. (2.32) and Eq. (2.33) have to be the same up to $\mathcal{O}\Big(\big(\alpha_s(Q^2)\big)^2\Big)$, we find

$$\mathbf{C}^{(1)}(z, \ln\big(\mu_R^2/Q^2\big)) = \mathbf{C}^{(1)}(z)\,. \tag{2.36}$$

Note that with this identification Eq. (2.32) and Eq. (2.33) (truncated to NLO) are not the same in practice, they are only perturbative equivalent. In any actual calculation with $\mu_R \neq Q$ also $\alpha_s(\mu_R^2)$ and $\alpha_s(Q^2)$ are different: they differ by an amount which is beyond our fixed-order framework and if that difference is "small" depends on the actual values of the strong coupling of the respective scales. However, with that we achieve the main purpose of this game: we generate some higher order terms, which we can then use to estimate MHOUs.

Starting from NNLO we see that the matching becomes non-trivial. In the case at hand we can insert Eq. (2.23) into Eq. (2.32) (with the choice $(\mu_R, \mu_{R,0}) \to (Q, \mu_R)$) and re-expand in powers of $\alpha_s(\mu_R^2)$ as requested. We find

$$\mathbf{C}^{(2)}(z, \ln\big(\mu_R^2/Q^2\big)) = \mathbf{C}^{(2)}(z) + \beta_0 \ln\big(\mu_R^2/Q^2\big)\mathbf{C}^{(1)}(z)\,, \tag{2.37}$$

so the scale-varied coefficient receives a correction from a lower-order coefficient function.

Let's take a step back and review the general features.

Conclusion 2.4: Renormalization scale variation

Scale-varied coefficient functions

- can always be computed a posteriori from their non-scale-varied counterparts
- are a linear combination of their lower order non-scale-varied counterparts
- are derived by re-expanding the strong coupling in the perturbative sum up to the required perturbative accuracy

We also know that the dependency on the scale variation $\ln\big(\mu_R^2/Q^2\big)$ must be a polynomial dependence since the running of strong coupling is a pure resummation of this logarithm. By re-expanding we mean a Taylor expansion of the strong coupling at the default scale, here $\alpha_s(Q)$, in terms of the value at the renormalization scale, $\alpha_s(\mu_R^2)$. For computing the strong coupling at any scale $\alpha_s(\mu_R^2)$ we must use required perturbative accuracy, but for deriving the scale-varied coefficient functions we only need one order less. In the special case of fully inclusive DIS, which has no dependence on the strong coupling at LO, we even need only two orders lower.

For the sake of correcting Eq. (1.3), we can improve it by writing

$$\sigma(x, Q^2, \mu_R^2) = \Big(\mathbf{C}^T(Q^2, \mu_R^2) \otimes \mathbf{f}(Q^2)\Big)(x)\,. \tag{2.38}$$

2.4.2. Factorization scale

As in Section 2.3.2 the general algorithm repeats for the factorization scale μ_F, which is the intrinsic scale of PDFs, except everything is more complicated. However, it is not only more complicated because PDFs are a set of function, but also because the PDF explicitly appears in Eq. (1.1). This implies, that the factorization scale dependence is not only be captured by the coefficient function, in contrast to the renormalization scale. In particular, the factorization scale captures collinear divergencies, which arise in diagrams, and re-attributes them to the PDF, thus there is a direct connection between the two. Recall our main goal here: we want to estimate MHOU by generating higher order terms and we simply have more freedom here to do so. In order to simplify the discussion we explicitly neglect renormalization scale variations for the moment.

The most common way to deal with factorization scale variation is to directly modify the coefficient functions, as in the case for Eq. (2.31), and we refer to this choice as "factorization scale variation scheme C" [49]. Specifically, we make a similar ansatz and write

$$\sigma(x, Q^2, \mu_F^2) = \Big(\mathbf{C}^T(Q^2, \mu_F^2) \otimes \mathbf{f}(\mu_F^2)\Big)(x) \tag{2.39}$$

where now μ_F has its intended place and the usual perturbative expansion of the coefficient function still holds, i.e.

$$\mathbf{C}(z, Q^2, \mu_F^2) = \sum_{j=0} \big(\alpha_s(Q^2)\big)^j \mathbf{C}^{(j)}(z, \ln\big(\mu_F^2/Q^2\big))\,. \tag{2.40}$$

However, in order to derive an explicit expression for $\mathbf{C}(z, Q^2, \mu_F^2)$ it is instructive to start from "factorization scale variation scheme B" [49]. In this scheme, we make a different choice and attribute the variation instead to the PDF:

$$\sigma(x, Q^2, \mu_F^2) = \Big(\mathbf{C}^T(Q^2) \otimes \mathbf{f}^{SV}(\mu_F^2, \ln(\mu_F^2/Q^2))\Big)(x) \tag{2.41}$$

Note that Eq. (2.41) only uses the central coefficient functions $\mathbf{C}^T(Q^2)$, but requires the use of a special PDF $\mathbf{f}^{SV}$. In a PDF fit, where full control over the PDF is required any way and partonic matrix elements (coefficient functions) come from a variety of places this scheme can be actually preferable.

In order to define $\mathbf{f}^{SV}$ we start by observing, that we can write Eq. (2.29) also as a convolution with a factorization scale operator $\mathbf{K}$

$$\mathbf{f}(\xi, \mu_F^2) = \big(\mathbf{K}(\mu_F^2 \leftarrow \mu_{F,0}^2) \otimes \mathbf{f}(\mu_{F,0})\big)(\xi) \tag{2.42}$$

where $\mathbf{K}$ admits a perturbative expansion

$$\mathbf{K}(y, \mu_F^2 \leftarrow \mu_{F,0}^2) = \sum_{j=0} \big(\alpha_s(\mu_{F,0}^2)\big)^j \mathbf{K}^{(j)}(y, \ln(\mu_F^2/\mu_{F,0}^2)) \tag{2.43}$$

and is given up to NLO accuracy by

$$\mathbf{K}(y, \mu_F^2 \leftarrow \mu_{F,0}^2) = \mathbb{1}\delta(1-y) + \alpha_s(\mu_{F,0}^2)\ln(\mu_F^2/\mu_{F,0}^2)\mathbf{P}^{(0)}(y) + \dots \tag{2.44}$$

Here, $\mathbb{1}$ refers to an identity operation in flavour space and $\delta(1-y)$ is the identity in the function convolution space as $\mathbf{K}$ is acting on both.

Note that $\mathbf{K}$ is a fixed order equivalent of the EKO $\mathbf{E}$, but the two are fundamentally different: the latter is a true solution to DGLAP, which must resum all necessary terms, but the former is only an approximation for a small amount of evolution. Specifically, $\mathbf{K}$ only is meaningful when we consider scale variations and in particular we use the interplay between $\mathbf{K}$ and $\mathbf{E}$ to generate the missing higher order terms we are looking for, because we get

$$\mathbf{f}(\xi, \mu_1^2) = \big(\mathbf{K}(\mu_1^2 \leftarrow \mu_2^2) \otimes \mathbf{E}(\mu_2^2 \leftarrow \mu_1^2) \otimes \mathbf{f}(\mu_1^2)\big)(\xi) + \dots \tag{2.45}$$

where ... denotes the missing terms. Inserting the equation back into Eq. (1.3) with $(\mu_1, \mu_2) \to (Q, \mu_F)$ we obtain the definition of our sought-after special PDF

$$\mathbf{f}^{SV}(\xi, \mu_F^2, \ln(\mu_F^2/Q^2)) = \left(\mathbf{K}(Q^2 \leftarrow \mu_F^2)\right) \otimes \mathbf{f}(\mu_F^2))(\xi) = \mathbf{f}(\xi, Q^2) + \ldots \tag{2.46}$$

where we apply Eq. (2.28) for the PDF.

We can now return to the definition of scheme C: we insert Eq. (2.46) into Eq. (2.41) and obtain

$$\sigma(x, Q^2, \mu_F^2) = \left[\mathbf{C}^T(Q^2) \otimes \left(\mathbf{K}(Q^2 \leftarrow \mu_F^2) \otimes \mathbf{f}^{SV}(\mu_F^2)\right)\right](x) \tag{2.47}$$

$$= \left[\left(\mathbf{C}^T(Q^2) \otimes \mathbf{K}(Q^2 \leftarrow \mu_F^2)\right) \otimes \mathbf{f}^{SV}(\mu_F^2)\right](x) \tag{2.48}$$

where we only used the associative property of the convolution. However, we can now re-expand terms in a different order. First, it is convenient to use the perturbative expansion of the strong coupling to rewrite $\mathbf{K}$ in powers of $\alpha_s(Q^2)$ (as opposed to $\alpha_s(\mu_F^2)$, Eq. (2.44)):

$$\mathbf{K}'(y, Q^2 \leftarrow \mu_F^2) = \sum_{j=0} \left(\alpha_s(Q^2)\right)^j \mathbf{K}'^{(j)}(y, \ln(\mu_F^2/Q^2)) \tag{2.49}$$

with

$$\mathbf{K}'(y, Q^2 \leftarrow \mu_F^2) = \left(\mathbf{K}(Q^2 \leftarrow \mu_F^2) \otimes \left[\mathbb{1} + \mathcal{O}(\alpha_s(Q^2))\right]\right)(y) \tag{2.50}$$

i.e. $\mathbf{K}'$ and $\mathbf{K}$ differ by higher order terms. Up to NLO accuracy we find from Eq. (2.44)

$$\mathbf{K}'(y, Q^2 \leftarrow \mu_F^2) = \mathbb{1}\delta(1-y) - \alpha_s(Q^2)\ln(\mu_F^2/Q^2)\mathbf{P}^{(0)}(y) + \ldots \tag{2.51}$$

Second, we use the definition of $\mathbf{C}(z, Q^2, \mu_F^2)$

$$\mathbf{C}^T(z, Q^2, \mu_F^2) = \left(\mathbf{C}^T(Q^2) \otimes \mathbf{K}'(Q^2 \leftarrow \mu_F^2)\right)(z) \tag{2.52}$$

which reorganizes the product of sums in Eq. (2.48) into a single sum, Eq. (2.40). Let's see this in practice.

As before, we start from LO accuracy and from Eq. (2.51) we find immediately

$$\mathbf{C}^{(0)}(z, \ln(\mu_F^2/Q^2)) = \mathbf{C}^{(0)}(z), \tag{2.53}$$

i.e., also here scale variations do not alter the LO coefficient function. However, unlike in the case for renormalization scale variations, factorization scale variations have always an impact starting from LO accuracy, because PDFs are present per-se in the factorization formula, Eq. (1.1). Specifically, in any actual calculation with $\mu_F \neq Q$ also $\mathbf{f}(\mu_F^2)$ and $\mathbf{f}(Q^2)$ are different: they differ by an amount which is beyond our fixed-order framework and if that difference is "small" depends on the actual values of the PDF of the respective scales. However, with that we achieve the main purpose of this game: we generate some higher order terms, which we can then use to estimate MHOUs. This is an intentional copy from Section 2.4.1.

Starting from NLO the matching becomes non-trivial and, e.g., at this order we find

$$\mathbf{C}^{(1),T}(z, \ln(\mu_F^2/Q^2)) = \mathbf{C}^{(1),T}(z) - \ln(\mu_F^2/Q^2)\left(\mathbf{C}^{(0),T} \otimes \mathbf{P}^{(0)}\right)(z)\,. \tag{2.54}$$

Always remember that for PDF operations, such as acting with the matrix of splitting functions $\mathbf{P}(y, \mu_F^2)$, non-commutativity is crucial, i.e., the flavor index dictated by the left-hand-side of Eq. (2.54) is matched by the second term on the right-hand-side by a vector-matrix multiplication (from the left). Note that starting from NNLO also explicit coefficients of the beta function start to appear, since the logarithm $\ln(\mu_F^2/Q^2)$ is coming directly from the running of the strong coupling, Eq. (2.22). In other words: PDF evolution resums logarithms through the running of the strong coupling.

In the following we assume scheme C as our default choice, thus conclusion 2.4 holds also for factorization scale variations and we can improve Eq. (1.3) with Eq. (2.39). The case of scheme B[3] is slightly more complicated, as explained above, but straightforward to obtain.

2.4.3. Combined variation

We recall, the renormalization and factorization scales are two independent scales, related to two different kind of physics. The former captures ultra-violet and the latter collinear divergencies. While we discuss them one at a time above, in practice we have to consider them simultaneously. In order to do so, we must first consider factorization scale variation,

[3] scheme A is doing yet something different - see Ref. [49]

which yields Eq. (2.39),

$$\sigma(x, Q^2, \mu_F^2, \mu_R^2 = Q^2) = \Big(\mathbf{C}^T(Q^2, \mu_F^2, \mu_R^2 = Q^2) \otimes \mathbf{f}(\mu_F^2)\Big)(x) \tag{2.55}$$

but now we make the renormalization scale dependence explicit and $\mathbf{C}(Q^2, \mu_F^2, \mu_R^2 = Q^2)$ corresponds to Eq. (2.40). Then in the second step, we apply the renormalization scale variation onto $\mathbf{C}(Q^2, \mu_F^2, \mu_R^2 = Q^2)$, which still depend on $\alpha_s(Q^2)$, and get

$$\sigma(x, Q^2, \mu_F^2, \mu_R^2) = \Big(\mathbf{C}^T(Q^2, \mu_F^2, \mu_R^2) \otimes \mathbf{f}(\mu_F^2)\Big)(x)\,. \tag{2.56}$$

We can make the dependence on the various scales in Eq. (2.56) more explicit by rewriting it as

$$\sigma(x, Q^2, \mu_F^2, \mu_R^2) = \Big(\mathbf{C}^T(\alpha_s(\mu_R^2), \ln\big(\mu_F^2/Q^2\big), \ln\big(\mu_R^2/Q^2\big)) \otimes \mathbf{f}(\mu_F^2)\Big)(x) \tag{2.57}$$

where, again, finally μ_F and μ_R fulfill their dedicated role. Note that in Eq. (2.57) two additional scales, $\mu_{F,0}$ and $\mu_{R,0}$ are left implicit and those eventually guarantee that it is a dimensionless equation.

Let's see this in practice and remember that in order to see renormalization scale logarithms we need to play at NNLO. We find

$$\begin{aligned}
&\mathbf{C}(z, \alpha_s(\mu_R^2), \ln\big(\mu_F^2/Q^2\big), \ln\big(\mu_R^2/Q^2\big)) \\
&= \mathbf{C}^{(0)}(z) \\
&\quad + \alpha_s(\mu_R^2)\Big[\mathbf{C}^{(1)}(z) + \mathbf{C}^{(1),F}(z)\ln\big(\mu_F^2/Q^2\big)\Big] \\
&\quad + \big(\alpha_s(\mu_R^2)\big)^2\Big[\mathbf{C}^{(2)}(z) + \mathbf{C}^{(2),F}(z)\ln\big(\mu_F^2/Q^2\big) + \mathbf{C}^{(2),R}(z)\ln\big(\mu_R^2/Q^2\big) \\
&\qquad\qquad \mathbf{C}^{(2),FF}(z)\ln^2(\mu_F^2/Q^2) + \mathbf{C}^{(2),FR}(z)\ln\big(\mu_F^2/Q^2\big)\ln\big(\mu_R^2/Q^2\big)\Big] + \dots
\end{aligned} \tag{2.58}$$

We see that the number of terms per perturbative order quickly grows as the respective algorithms collect more of the full behavior. The appearance of $\mathbf{C}^{(2),FR}$ also demonstrate how the two variations become intertwined.

2.4.4. Discussion

The renormalization and factorization scale are remnants of the necessary regularization of UV and collinear divergencies. Although, scale variations are a genuine pQCD property they have some features, which questions their usefulness for estimating MHOUs.

When introducing the scales, we are required to set them to “a typical scale of the process”, which makes them inherently arbitrary from the start before doing any variation. In fully inclusive DIS there is a single scale available, the virtuality Q^2, and thus the common choice is $\mu_F^2 = Q^2 = \mu_R^2$, but any multiplicative there of could work just as well[4]. If we want to use scale variations to estimate MHOU, we need to make a further arbitrary choice about which range we want to use for the respective scales and if we want to correlate them. As we see in conclusion 2.4 the scale-varied coefficient functions are always a linear combination of the existing ones, i.e. they can not introduce new channels (new PDF combinations). However, it is often straightforward to show that a new channel should appear although its size is unknown and scale variations are unable to predict this. An example for such a scenario in fully inclusive DIS can be seen at NLO: starting from NLO the gluon can participate in the scattering, but the LO only contains quark PDFs. In other cases, we know that higher orders must contain certain kinematic logs, e.g. from particles radiating at small and/or large transverse momentum or in the soft limit and in fully inclusive DIS the latter is the case[5]. Also those corrections can (in many cases) not be predicted by scale variations.

Scale variations also provide no statistical interpretation, which is in stark contrast to experimental uncertainties, where we usually assume Gaussian uncertainties and thus the provided error has a clear interpretation. However, strictly speaking this only applies to statistical errors but not systematic errors and the question which of the dominates depends very much on the specific measurement.

Despite all these (potential) problems, scale variations have a solid foundation in QFT and can be applied (straightforwardly) to any pQCD process, which makes them an easy tool to use MHOU. In order to make a faithful statement, we can then simply say: when we compute this process with this specific choice of scale, we obtain this number; if we vary that scale in this range, we obtain this range of numbers.

[4] For example CT18X [60] makes a different choice.

[5] More mathematically speaking: soft gluon resummation is a double logarithmic resummation $\tilde{c}_q^{(j)} \sim \alpha_s^j \ln^{2j}(N)$ for $N \to \infty$ in Mellin space, Eq. (2.26).

2.4.5. Implementation

Let us return now to more practical consequences. As we see above, renormalization scale variations are just a scalar operation, but factorization scale variations are more complicated and it is worth spelling out their explicit construction to see what is actually happening.

The factorization scale variation is directly related to the PDFs themselves and, in particular, eventually, they derive from Eq. (2.25). However, the splitting functions $\mathbf{P}$ are encoding how partons couple among each other, which is through the gluon[6] of course. This is in contrast to DIS, where instead the coefficient functions are encoding how partons couple to the electro-weak vector boson, which is completely different, see conclusion 1.1.

As usual, at LO accuracy the problem does not appear yet, because there are no scale-varied coefficient functions, see Eq. (2.53), but starting from NLO accuracy we need to worry. In particular, we know that the gluon couples the same to all quarks, thus, if we assume four flavors (see Section 2.5) for the moment and combine them in the following way [61][7],

$$\begin{aligned} f_{\Sigma_\Delta}(x,Q^2) = f_u(x,Q^2) + f_{\bar{u}}(x,Q^2) + f_c(x,Q^2) + f_{\bar{c}}(x,Q^2) \\ - f_d(x,Q^2) - f_{\bar{d}}(x,Q^2) - f_s(x,Q^2) - f_{\bar{s}}(x,Q^2) \end{aligned} \tag{2.59}$$

we know that this combination does not interact with the gluon. In order to see more clearly what happens we need to go back to the vector notation from Eq. (1.3). We repeat: we denote objects, which have a non-trivial dependence on the parton flavors, with bold font. This applies to vectors in flavor space, e.g. coefficient functions $C_j \to \mathbf{C}$ or PDFs $f_j \to \mathbf{f}$, but also to matrices, e.g. splitting functions $P_{jk} \to \mathbf{P}$ or the factorization scale operator $K_{jl} \to \mathbf{K}$. By introducing the vector notation we hide any reference to a given basis in the flavor space, which now becomes relevant (see also Section 2.5).

For computing LO and NLO DIS coefficient functions it is convenient to consider each quark individually and we can find easily Eqs. (1.4) and (1.16). We refer to this choice of basis as the flavor basis. However, for the discussion in Section 2.3.2 and in particular for dealing with splitting functions, $\mathbf{P}$, a different basis is more convenient: the evolution basis. The two choices are both orthogonal bases, i.e. they are simply related

[6] for quark-photon interaction see Ref. [61]
[7] for the more general case see Ref. [61]

by a rotation[8] in flavor space. Specifically, $\mathbf{P}$ has a much simpler representation if one of the basis vectors is the singlet

$$\begin{aligned} f_\Sigma(x,Q^2) = f_u(x,Q^2) + f_{\bar{u}}(x,Q^2) + f_c(x,Q^2) + f_{\bar{c}}(x,Q^2) \\ + f_d(x,Q^2) + f_{\bar{d}}(x,Q^2) + f_s(x,Q^2) + f_{\bar{s}}(x,Q^2) \end{aligned} \tag{2.60}$$

and all other basis vectors can be chosen such that they do not couple to the gluon, e.g. Eq. (2.59).

Let's see this in practice for LO photon DIS: we start from Eq. (1.16), which is written in flavor basis, i.e. for each given quark flavor and the gluon. Inserting Eq. (1.16) into Eq. (1.1) yields Eq. (1.4) and we find explicitly

$$\begin{aligned} \sigma(x,Q^2) = e_d^2\big(f_d(x,Q^2) + f_{\bar{d}}(x,Q^2) + f_s(x,Q^2) + f_{\bar{s}}(x,Q^2)\big) \\ + e_u^2\big(f_u(x,Q^2) + f_{\bar{u}}(x,Q^2) + f_c(x,Q^2) + f_{\bar{c}}(x,Q^2)\big) \end{aligned} \tag{2.61}$$

Using Eqs. (2.59) and (2.60) we can rewrite this as

$$\sigma(x,Q^2) = \left(e_d^2 + e_u^2\right) f_\Sigma(x,Q^2) + \left(e_u^2 - e_d^2\right) f_{\Sigma_\Delta}(x,Q^2) \tag{2.62}$$

or equivalently

$$C_\Sigma^{(0)}(z) = (e_d^2 + e_u^2)\delta(1-z),\quad C_{\Sigma_\Delta}^{(0)}(z) = (e_u^2 - e_d^2)\delta(1-z),\quad C_j(z) = 0\,\forall j \neq \{\Sigma, \Sigma_\Delta\} \tag{2.63}$$

We stress that Eqs. (1.16) and (2.63) are two equivalent representation of the coefficient function $\mathbf{C}^{(0)}$ using two different bases.

Moving to NLO, we have the same structure in the quark sector, Eq. (1.17), and we encounter for the first time the gluon channel, Eq. (1.19). Thus, when we want to execute Eq. (2.54), we need to consider three directions in flavor space: the gluon, the singlet f_Σ and Eq. (2.59). The preferred basis for $\mathbf{P}$ is manifest by the following statements,

$$\mathrm{P}_{\Sigma\Sigma_\Delta}(y) = \mathrm{P}_{g\Sigma_\Delta}(y) = \mathrm{P}_{\Sigma_\Delta\Sigma}(y) = \mathrm{P}_{\Sigma_\Delta g}(y) = 0 \tag{2.64}$$

[8] actually, it is a rotation and a scaling, i.e. the evolution basis is not orthonormal, which may cause some trouble

which would not occur in flavor basis. Eventually, Eq. (2.54) becomes explicitly

$$C^{(1)}_{\Sigma_\Delta}(z, \ln(\mu_F^2/Q^2)) = C^{(1)}_{\Sigma_\Delta}(z) - \ln(\mu_F^2/Q^2)\left(C^{(0)}_{\Sigma_\Delta} \otimes \mathrm{P}^{(0)}_{\Sigma_\Delta\Sigma_\Delta}\right)(z) \tag{2.65}$$

$$C^{(1)}_{\Sigma}(z, \ln(\mu_F^2/Q^2)) = C^{(1)}_{\Sigma}(z) - \ln(\mu_F^2/Q^2)\left(C^{(0)}_{\Sigma} \otimes \mathrm{P}^{(0)}_{\Sigma\Sigma}\right)(z) \tag{2.66}$$

$$C^{(1)}_{g}(z, \ln(\mu_F^2/Q^2)) = C^{(1)}_{g}(z) - \ln(\mu_F^2/Q^2)\left(C^{(0)}_{\Sigma} \otimes \mathrm{P}^{(0)}_{\Sigma g}\right)(z) \tag{2.67}$$

where the required splitting functions can be found in the literature [4–6].

Actually, also LO splitting functions $\mathbf{P}^{(0)}$ obey symmetry relations, in particular all quarks couple identically to the gluon and it is not possible (yet) to change the quark flavor. This implies $\mathrm{P}^{(0)}_{\Sigma_\Delta\Sigma_\Delta}(y) = \mathrm{P}^{(0)}_{\Sigma\Sigma}(y) = \mathrm{P}^{(0)}_{qq}(y)$ and thus Eq. (2.54) can also be executed straightforwardly in flavor space. However, this symmetry breaks down starting from NLO splitting functions $\mathbf{P}^{(1)}$, which would be required for NNLO DIS cross sections.

2.4.6. Interplay with previous content

- Renormalization scale variation is a reshuffling of lower order coefficient functions and factorization scale variation adds convolutions between splitting functions and lower order coefficient functions. This implies that the mathematical complexity remains the same or in other words: we never encounter new mathematical objects in scale variations; see Section 1.3.
- Although scale-varied coefficient functions can always be computed a posteriori, see conclusion 2.4, in practice it usually makes sense to tabulate them into interpolation grids as well, see Section 2.2.3. These additional coefficient functions can be easily accommodated when we generalize our concept of "order": it can not only represent the perturbative order per se, but also additional powers of renormalization or factorization logarithms: $\alpha_s^k(\mu_R^2)\ln^m(\mu_R^2/Q^2)\ln^n(\mu_R^2/Q^2)$.
- The necessary projection onto a singlet component is particularly tricky for CC DIS, see Section 2.1. Recall that the $W^\pm$ boson only couple to specific PDFs and thus if we want to project the coefficient functions into evolution basis we need the full basis (and not just Eqs. (2.59) and (2.60)).
- Note that the coupling structure is always retained, i.e.

$$C^{(1)}_{g}(z) \sim e_u^2 + e_d^2 \sim C^{(1)}_{g}(z, \ln(\mu_F^2/Q^2)) \tag{2.68}$$

This implies that a given "channel" in an interpolation grid (Section 2.2.3) has a unique coupling, though recall that some channels only appear at high enough perturbative order.

2.5. Heavy quarks

When we compute LO DIS from Fig. 1.3a we say it is easy to identify the respective coefficient function, Eq. (1.16). However, what exactly do we mean with the "$\forall q$"? Which quarks are we exactly talking about? This is directly linked to the flavor space to which we hint when using bold font. However, unlike the problem we discuss in Section 2.4.5 about different bases (flavor basis vs. evolution basis), here we are more directly concerned what the flavor space is in the first place.

2.5.1. Flavor space

We define the flavor space $\mathcal{F}$ as the vector space spanned by all quark flavors and the gluon. We consider the naive flavor basis as our canonical basis and write

$$\mathcal{F} = \mathcal{F}_{fl} = \text{span}(g, u, \bar{u}, d, \bar{d}, s, \bar{s}, c, \bar{c}, b, \bar{b}, t, \bar{t})\,. \tag{2.69}$$

We have a 13 dimensional vector space, $\dim \mathcal{F} = 13$[9], consisting of the gluon, six quarks, and six anti-quarks. Our bold vectors are an element in that space, e.g. $\mathbf{C} \in \mathcal{F}$, or more plainly,

$$\mathbf{C}^T \sim (C_g, C_u, C_{\bar{u}}, C_d, C_{\bar{d}}, C_s, C_{\bar{s}}, C_c, C_{\bar{c}}, C_b, C_{\bar{b}}, C_t, C_{\bar{t}}) \tag{2.70}$$

and, as said in Section 2.4.5, any other basis works just as well. The same applies also to the PDF $\mathbf{f}$, which is the other vector in Eq. (1.3), and any operator, e.g. splitting functions $\mathbf{P}$, just lives in the product space, $\mathbf{P} \in \mathcal{F} \otimes \mathcal{F}$.

The question we need to address here is: which quarks exactly can participate in Eqs. (1.4) and (1.16)? We refer to this number as the number of active quarks n_f. This dependency is completely hidden in Eq. (1.1), but, as we demonstrate in the following, is

[9] one can add the photon and make it 14 dimensional [61]

a major concern for both coefficient functions and PDF. Note that the gluon is always considered active and contributes to all diagrams always.

The number of flavors n_f is arbitrary per se and there is no absolutely correct answer, since there is no theory argument, which would fix this choice. However, in practice for a given observable $\sigma(x, Q^2)$ we can consider, which options are phenomenologically reasonable and if some option would yield unstable results.

For the sake of the argument, let's consider a fully inclusive DIS cross section $\sigma(x, Q^2 = 25\,\mathrm{GeV}^2)$, so we probe the probe the proton at $\mu_F = Q = 5\,\mathrm{GeV}$. We can now consider the different quark masses as a guideline: on the one side the masses of up, down, and strange are significantly below the non-perturbative regime of QCD $m_u, m_d, m_s \ll \Lambda_{\mathrm{QCD}} \sim 1\,\mathrm{GeV}$ and we can safely assume we can find them in the proton. On the other side the mass of the top is significantly above the probed scale, $m_t \gg Q$, thus it is very unlikely to find it in the proton and we can neglect it. We can shut down such contributions by setting $f_t(\xi, Q^2) = 0 = f_{\bar{t}}(\xi, Q^2)$ and this way the top quark does not contribute to Eq. (1.1). In addition, we can also define $C_t(z, Q^2) = 0 = C_{\bar{t}}(z, Q^2)$ to explicitly say that we don't let the top contribute.

Now, we consider the two remaining quarks, charm and bottom, for which maybe there is no clear answer. First, for the charm quark we have $m_c \sim 1.5\,\mathrm{GeV}$ and we can assume reasonably that it can contribute to DIS. On the other side, for the bottom quark we have $m_b \sim 4.5\,\mathrm{GeV}$ and maybe we can both allow or deny it. The specific choice on which quarks we allow to participate and in which way they do so is referred to as flavor number scheme.

2.5.2. ZM-VFNS

One very popular choice is the Zero-Mass Variable Flavor Number Scheme (ZM-VFNS), which is a telling name and which we have silently adopted in the previous parts. Formally, we can define it in the following way: first, whenever the probed scale Q crosses a quark mass that quark becomes active; second; in coefficient functions all active quarks are considered massless and all non-active quarks are considered infinite massive. The infinite mass ensures $f_H(\xi, Q^2) = 0 = f_{\bar{H}}(\xi, Q^2)$ and $C_H(z, Q^2) = 0 = C_{\bar{H}}(z, Q^2)$ for all non-active quarks H and the zero mass ensures that all formulas in Section 1.3 hold. If we denote the new dependency on the number of active flavors n_f by a superscript (n_f)

we can define the ZM-VFNS in the following way

$$\sigma^{\text{ZM-VFNS}}(x,Q^2)=\begin{cases}\sigma^{(3)}(x,Q^2) & \text{if } Q^2<m_c^2\\ \sigma^{(4)}(x,Q^2) & \text{if } m_c^2\le Q^2<m_b^2\\ \sigma^{(5)}(x,Q^2) & \text{if } m_b^2\le Q^2<m_t^2\\ \sigma^{(6)}(x,Q^2) & \text{if } Q^2\ge m_t^2\end{cases} \tag{2.71}$$

where

$$\sigma^{(n_f)}(x,Q^2)=\left(\mathbf{C}^{T,(n_f)}(Q^2)\otimes \mathbf{f}^{(n_f)}(Q^2)\right)(x) \tag{2.72}$$

i.e. each contribution has its own factorization formula and we have, e.g.,

$$\begin{matrix}C_H^{(3)}(z,Q^2)=0=C_{\bar{H}}^{(3)}(z,Q^2)\\ f_H^{(3)}(\xi,Q^2)=0=f_{\bar{H}}^{(3)}(\xi,Q^2)\end{matrix}\qquad H\in\{c,\bar{c},b,\bar{b},t,\bar{t}\} \tag{2.73}$$

and similar for the others. Specifically, we can improve Eq. (1.16) by writing

$$C_q^{(0),(3)}(z)=e_q^2\delta(1-z)\qquad q\in\{c,\bar{c},b,\bar{b},t,\bar{t}\} \tag{2.74}$$
$$C_H^{(0),(3)}(z)=0\qquad H\in\{c,\bar{c},b,\bar{b},t,\bar{t}\} \tag{2.75}$$
$$C_g^{(0),(3)}(z)=0 \tag{2.76}$$

and we finally give full details on the available flavor space $\mathcal{F}$.

Note that it does not make sense to consider less then three quarks, since $m_u, m_d, m_s \ll \Lambda_{\text{QCD}} \sim 1\,\text{GeV}$ and pQCD only works for $Q > \Lambda_{\text{QCD}}$. All quarks are considered with their actual mass for the decision if they are in or out, Eq. (2.71), but then after that their mass is discarded when computing $\mathbf{C}^{(n_f)}(z,Q^2)$. Thus, the VFNS indicates that the number of active quarks n_f depends on the probed scale Q, i.e. it is variable, and the ZM explains how to compute diagrams. However, once we have an actual scale in hand, n_f is also fixed.

> Conclusion 2.5: ZM-VFNS
>
> In the ZM-VFNS
>
> - the number of active quarks n_f depends on the probed scale Q^2
> - contain the diagrams to compute coefficient functions only massless quarks

2.5.3. FFNS

For the number of active quarks we can make another simple choice: fix n_f once and forever - the Fixed Flavor Number Scheme (FFNS). Here, the probed scale Q does not matter, but we use the same coefficient functions for any scale. Actually, saying FFNS is not sufficient, we still have some choices and the most obvious is n_f.

However, we can also improve with respect to other parts: while it is significantly easier to compute diagrams with only massless quarks, we can compute them also with massive quarks. For simplicity we assume in the following we have exactly one[10] massive quark with mass m^2. Moreover, while we keep the notation general you can always consider as a concrete example $n_f = 3$ and thus $m^2 = m_c^2$. We recall two important features of the propagator of a massive quark,

$$D_q(p) \sim \frac{p_\mu \gamma^\mu + m\mathbb{1}}{p^2 - m^2}\,; \tag{2.77}$$

first, the mass appears explicitly in the numerator, which makes computing Dirac traces significantly more lengthy, and, second, the mass appears in the denominator, which shields the propagator from a collinear pole. When we consider a finite mass we eventually have three different scales in our diagrams: the total energy of the colliding partons $\hat{s}$, the photon virtuality Q^2, and the quark mass m^2. This means that unlike the massless case discussed so far, the coefficient functions now depend on two scales, for example the partonic momentum fraction $z = Q^2/(\hat{s} + Q^2)$ and $\xi_q = Q^2/m^2$ or any other bijective combination[11]. This in turn implies that the available mathematical function space is much bigger, which makes the computation of these coefficient functions $\mathbf{C}(z, \xi_q, Q^2)$ significantly more complex.

[10] the case of two massive quarks is e.g. discussed in Ref. [62]

[11] For example, for NC DIS on can show [3] that $\chi = \frac{1-\sqrt{1-4m^2/s}}{1+\sqrt{1-4m^2/s}}$ and $\chi_q = \frac{\sqrt{1+4m^2/Q^2}-1}{\sqrt{1+4m^2/Q^2}+1}$ are good combinations.

However, we focus here on the new dependency on ξ_q, which captures the new features. First, we consider the case $Q^2 \gg m^2$ or $\xi_q \to \infty$ where the denominator of Eq. (2.77) plays the important role. Indeed, since it is shielding the propagator from a collinear divergency, we encounter some signals thereof when we put $m^2 \to 0$. Specifically, we can rewrite the coefficient functions in this limit,

$$\lim_{Q^2/m^2 \to \infty} \mathbf{C}^{(n_f),T}(Q^2, m^2) = \mathbf{C}^{(n_f+1),T}(Q^2)\mathbf{A}^{(n_f+1)}(Q^2, m^2) \tag{2.78}$$

i.e. we can connect the coefficient functions using two neighboring schemes and a new, universal operator $\mathbf{A}$, which admits a perturbative series

$$\mathbf{A}^{(n_f+1)}(Q^2, m^2) = \sum_{j=0} \left(\alpha_s(Q^2)\right)^j \mathbf{A}^{(j),(n_f+1)}(\ln(Q^2/m^2)) \,. \tag{2.79}$$

We discuss more details of this new operator in a moment, but the important thing to stress here is the universality. This universality derives from the universality of PDFs, which makes the operator closely connected to collinear splitting functions as we expect. Moreover, Eq. (2.78) puts strong constraints on the computation of any coefficient function[12] once the universal operator $\mathbf{A}$ is known at the requested accuracy. Note that the coefficient functions in Eq. (2.78) depend on different sets of variables, since we consider the "+1" quark massless in the $(n_f + 1)$ scheme. Thus, as expected, the full mass dependence on the right-hand-side of Eq. (2.78) is carried by the universal operator $\mathbf{A}$ and it is a pure, actual polynomial, dependency on $\ln\left(Q^2/m^2\right)$. So, the remnant of the collinear divergency for a massive quark is a logarithm $\ln\left(Q^2/m^2\right)$ as we may have guessed from the beginning.

Next, we consider the case $Q^2 \sim m^2$ where the nominator of Eq. (2.77) plays the important role. While we conclude above that the FFNS coefficient functions contain logarithms $\ln\left(Q^2/m^2\right)$ they also naturally contain terms which are relatively suppressed to them. Specifically, if we expand the FFNS coefficient function around $Q^2 \sim m^2$ we obtain

$$\mathbf{C}^{(n_f)}(Q^2, m^2) = \mathbf{C}^{(n_f)}_{thr,0} + \mathbf{C}^{(n_f)}_{thr,1}\left(\frac{Q^2}{m^2}\right) + \mathbf{C}^{(n_f)}_{thr,2}\left(\frac{Q^2}{m^2}\right)^2 + \dots \tag{2.80}$$

with some coefficients $\mathbf{C}^{(n_f)}_{thr,k}$. Note that the equation is to be understood as a Taylor expansion at the requested point as in practice the actual dependency on ξ_q is very

[12] or any partonic matrix elements as a similar equations hold, e.g., in hadron-hadron collisions

intricate[13]. However, it is clear that in the region $Q^2 \sim m^2$ these terms may not be small and thus neglecting such contributions may yield wrong results. That is the precisely the advantage of FFNS over ZM-VFNS: while in the latter we explicitly deny any massive quark in the diagrams, the former allows them to be included in the sum over diagrams.

Conclusion 2.6: FFNS

In the FFNS

- massive quarks are allowed in the computation of diagrams for coefficient functions
- we counter universal logarithms $\ln\left(Q^2/m^2\right)$ for $Q^2 \to m^2$ due to collinear divergencies

But if the FFNS calculation is actually better than ZM-VFNS, why don't we always use the former? Before answering the question, we need to return to the new operator $\mathbf{A}$.

2.5.4. Matching schemes

The number of active quarks is not only relevant for coefficient functions, but, in fact, for the whole pQCD program and in particular for the strong coupling α_s and PDFs $\mathbf{f}$. As we see from Eq. (2.72) also PDFs depend on the number of flavors, e.g. $\mathbf{f}^{(3)}(\xi, \mu_F^2)$, and we can improve Eq. (1.15) with

$$\mathbf{C}^{(n_f)}(z, Q^2, m^2) = \sum_{k=0} \left(\alpha_s^{(n_f)}(Q^2)\right)^k \mathbf{C}^{(k),(n_f)}(z, \xi_q) \tag{2.81}$$

where we now also explicitly denote the flavor dependence of the strong coupling. Actually, from Eq. (2.20) we already known that the beta coefficients depend explicitly on n_f and thus α_s must depend on it as well. The same applies to PDFs since also splitting functions carry an explicit n_f dependence and all appearances above, indeed, correspond to $\mathbf{P}^{(n_f)}$.

The question is: how are those different schemes, e.g. $\alpha_s^{(3)}$ and $\alpha_s^{(4)}$, connected with each other? For coefficient functions the difference originates from the different diagrams which are included into the calculation of the coefficient functions, $\mathbf{C}^{(3)}$ and $\mathbf{C}^{(4)}$, but what about the strong coupling α_s and the PDFs $\mathbf{f}$? In order to answer this question

[13] for example via χ_q

it is important to return to the basic principles: we know that only cross sections σ are physical objects for which we can derive physical properties, e.g. that they must be positive. In contrast, the strong coupling α_s, PDFs $\mathbf{f}$, and coefficient functions $\mathbf{C}$ are not physical objects, which can not be measured directly in an experiment and which only are defined within our framework of pQCD. The concept of having different number of active flavors n_f is a theoretical idea, which we employ to make a consistent approximation in our calculation, but is irrelevant to nature. In pQCD we can still derive certain rules for our relevant objects in order to obtain a consistent theory, which we can then use to make predictions.

The important physical properties, which we need here is consistency and perturbative equivalence. Specifically, we require that the cross section in two different scheme are perturbatively equivalent, i.e.

$$\sigma^{(n_f)}(x,Q^2) = \sigma^{(n_f+1)}(x,Q^2)\left[1+\mathcal{O}\left(\alpha_s^{(n_f+1)}(Q^2)\right)\right]\,, \tag{2.82}$$

where the two cross section are consistently computed in their respective flavor number scheme. In particular, they use their own coefficient functions, their own strong coupling, and their own PDFs. However, since the coefficient functions, the beta coefficients, and the splitting functions are computed from the respective relevant diagrams this makes the condition quite restrictive. Moreover, if we recall Section 2.3 we know that both, α_s and $\mathbf{f}$, perform a resummation, so Eq. (2.82) is to be understood in a logarithmic sense and thus, in practice, it is convenient to consider Eq. (2.82) in the limit $Q^2 \to \infty$.

One can show [62] that the connection between two different schemes can be done by a universal matching procedure for the strong coupling and the PDFs. Specifically, for the strong coupling we find

$$\alpha_s^{(n_f+1)}(\mu_R^2) = \zeta^{(n_f+1)}(\mu_R^2, m^2)\alpha_s^{(n_f)}(\mu_R^2) \tag{2.83}$$

with a universal matching operator ζ, which admits a perturbative series and has a polynomial dependence on $\ln\left(\mu_R^2/m^2\right)$. Analogously, for the PDFs we find

$$\mathbf{f}^{(n_f+1)}(\xi,\mu_F^2) = \left(\mathbf{A}^{(n_f+1)}(\mu_F^2,m^2)\otimes\mathbf{f}^{(n_f)}(\mu_F^2)\right)(\xi) \tag{2.84}$$

with a universal matching operator $\mathbf{A}$, which is also used in Eq. (2.78). The scheme change, Eq. (2.83) and Eq. (2.84), can be done at any scale, but, following Eq. (2.71), is conventionally done at the mass of the heavy quark, $\mu_R^2 = m^2 = \mu_F^2$.

Let's see this in practice: consider a cross section $\sigma(x,Q^2)$ with $Q^2 > m^2$. As we say above the number of active quarks is a free choice, we can compute the cross section either in the n_f scheme, Eq. (2.72), or in the (n_f+1) scheme,

$$\sigma^{(n_f+1)}(x,Q^2) = \left(\mathbf{C}^{T,(n_f+1)}(Q^2) \otimes \mathbf{f}^{(n_f+1)}(Q^2)\right)(x)\,. \tag{2.85}$$

Inserting Eq. (2.84) into Eq. (2.85) we obtain

$$\sigma^{(n_f+1)}(x,Q^2) = \left(\mathbf{C}^{T,(n_f+1)}(Q^2) \otimes \mathbf{A}^{(n_f+1)}(Q^2,m^2) \otimes \mathbf{f}^{(n_f)}(Q^2)\right)(x) \tag{2.86}$$

and if Eq. (2.78) holds, the two expressions, Eq. (2.72) and Eq. (2.85), are actually perturbatively equivalent. Recall that Eq. (2.78) only holds in the $Q^2 \gg m^2$ limit and so in practice for finite Q^2 the two expression are only perturbatively equivalent.

Equations (2.83) and (2.84) imply that the two quantities are not continuous across the matching, but they differ by the application of the matching operator. Actually, at low enough perturbative order the matching operator, ζ and $\mathbf{A}$, are trivial (e.g. $\zeta(\mu_R^2,m^2) = 1 + \mathcal{O}\big(\alpha_s(\mu_R^2)\big)$) and the discontinuity is a higher-order effect. The strong coupling, the PDFs, and the coefficient function, who are on their own all discontinuous, conspire together in such a way that the cross section is continuous at the requested accuracy, Eq. (2.82). This maps directly to the statement from Section 1.1.3 that the cross section is a measurable object, which obviously must be continuous, but the others not.

2.5.5. FONLL as an example of a GM-VFNS

Let's return to the question on why we not always consider the FFNS. Looking to Eq. (2.86), we note that we have matched the PDF at Q^2, which is what we have needed there to make the argument of consistency work. However, we also say that the usual thing instead is to match at the heavy quark mass m^2, so let's see how that plays out.

Starting from Eq. (2.85), we find

$$\sigma^{(n_f+1)}(x, Q^2)$$

$$= \left(\mathbf{C}^{T,(n_f+1)}(Q^2) \otimes \mathbf{f}^{(n_f+1)}(Q^2)\right)(x) \tag{2.87}$$

$$= \left(\mathbf{C}^{T,(n_f+1)}(Q^2) \otimes \mathbf{E}^{(n_f+1)}(Q^2 \leftarrow m^2) \otimes \mathbf{f}^{(n_f+1)}(m^2)\right)(x) \tag{2.88}$$

$$= \left(\mathbf{C}^{T,(n_f+1)}(Q^2) \otimes \mathbf{E}^{(n_f+1)}(Q^2 \leftarrow m^2) \otimes \mathbf{A}^{(n_f+1)}(m^2, m^2) \otimes \mathbf{f}^{(n_f)}(m^2)\right)(x) \tag{2.89}$$

where in the second equation we have inserted the EKO in the (n_f+1) scheme, $\mathbf{E}^{(n_f+1)}$, from Eq. (2.28) and then in the third equation the matching, Eq. (2.84), which is now in the customary place. We can massage Eq. (2.72) in a similar way and find

$$\sigma^{(n_f)}(x, Q^2) = \left(\mathbf{C}^{T,(n_f)}(Q^2) \otimes \mathbf{f}^{(n_f)}(Q^2)\right)(x) \tag{2.90}$$

$$= \left(\mathbf{C}^{T,(n_f)}(Q^2) \otimes \mathbf{E}^{(n_f)}(Q^2 \leftarrow m^2) \otimes \mathbf{f}^{(n_f)}(m^2)\right)(x) \tag{2.91}$$

with the EKO now in the n_f scheme, $\mathbf{E}^{(n_f)}$.

As we discuss in Section 2.3 the EKO, $\mathbf{E}$, is performing a resummation and in this particular case a resummation of the logarithm $\log\left(Q^2/m^2\right)$ as can be seen from the arguments in Eqs. (2.89) and (2.91). This is true in either scheme, but the amount of logarithms, meaning the prefactor that is resummed along side, is different. The resummation is directly proportional to the splitting functions, Eq. (2.29), which is different between the two scheme. In particular, in the (n_f) scheme we are resumming the collinear logs of n_f light quarks, but in the (n_f+1) scheme we are resumming the collinear logs of (n_f+1) light quarks. Thus the latter contains effectively more logarithms.

But then how could we find an equivalence between the two schemes in Section 2.5.4? Recall that the equivalence is valid for cross sections, where coefficient functions explicitly enter, Eq. (2.72). Indeed from Section 2.5.3 we know that $\mathbf{C}^{(n_f)}(z, Q^2, m^2)$ contains logarithms $\ln\left(Q^2/m^2\right)$, which are precisely the ones we are hunting here. However, the big difference is that $\mathbf{C}^{(n_f)}(z, Q^2, m^2)$ only contains a finite amount of logarithms, where as $\mathbf{E}^{(n_f+1)}$ is resumming those logarithms. This means that for $Q^2 \gg m^2$ FFNS is indeed a bad choice as the logarithms $\ln\left(Q^2/m^2\right)$ are big and better be resummed. Instead any power-suppressed terms, Eq. (2.80), which are important in the region $Q^2 \sim m^2$ and made FFNS preferable over ZM-VFGNS, are irrelevant in this region.

Conclusion 2.7: GM-VFNS

A General Mass-Variable Flavor Number Scheme (GM-VFNS) must approximate a cross section $\sigma(x, Q^2)$

- in the region $Q^2 \sim m^2$ with the FFNS result $\sigma^{(n_f)}(x, Q^2, m^2)$
- in the region $Q^2 \gg m^2$ with the resummed ZM-VFNS result $\sigma^{(n_f+1)}(x, Q^2)$

Conclusion 2.7 gives no exact prescription on how the approximation has to be done and, indeed, there is some freedom on how to do this. Different PDF fitting groups make different choices [62–70] and as an exemplary case we consider here FONLL [62,68], which has a simple idea: in order to match the two regions, we simply add the two expressions together and subtract any overlap. Let's consider the concrete case of charm:

$$\sigma^{\text{FONLL}}(x, Q^2, m_c^2) = \sigma^{(3)}(x, Q^2, m_c^2) + \sigma^{(4)}(x, Q^2) - \sigma^{(3\cap 4)}(x, Q^2, m_c^2)\,. \tag{2.92}$$

Here,

- $\sigma^{(3)}(x, Q^2, m_c^2)$ represents the massive FFNS result, which contains power-like Q^2/m_c^2 terms and a finite amount of $\ln\left(Q^2/m_c^2\right)$ associated with the charm
- $\sigma^{(4)}(x, Q^2)$ represents the massless ZM-VFNS result, which resums all logarithms $\ln\left(Q^2/m_c^2\right)$, including the ones associated with the charm
- $\sigma^{(3\cap 4)}(x, Q^2, m_c^2)$ collects all terms, which appear in both expressions simultaneously, and for which we have borrowed the notation from set notation

One can show [62] that the overlap contribution can be written as

$$\sigma^{(3\cap 4)}(x, Q^2, m_c^2) = \left(\mathbf{C}^{(3\cap 4)}(Q^2, m^2) \otimes \mathbf{f}^{(3)}(Q^2)\right)(x) \tag{2.93}$$

where the corresponding coefficient function is given by

$$\mathbf{C}^{(3\cap 4),T}(z, Q^2, m^2) = \left(\mathbf{C}^{(4),T}(Q^2) \otimes \mathbf{A}^{(4)}(Q^2, m^2)\right)(x)\,, \tag{2.94}$$

i.e. the right-hand-side of Eq. (2.78). It is immediately clear that in the limit $Q^2 \gg m^2$ the coefficient functions $\mathbf{C}^{(3),T}$ and $\mathbf{C}^{(3\cap 4),T}$ become identical and, thus, in Eq. (2.92) only $\sigma^{(4)}$ contributes as requested by conclusion 2.7. On the other side, since $\sigma^{(3\cap 4)}$ contains exactly the logarithmic terms of $\sigma^{(3)}$ and those are also contained in $\sigma^{(4)}$, which actually only contains logarithms, we also match the $Q^2 \sim m^2$ region. The yet additional logarithms in $\sigma^{(4)}$, which are resummed there, are small in this region and a part of the approximation FONLL takes.

2.5.6. Interplay with previous content

- The relevant parameter in conclusion 2.7 is the scale Q^2, which, as we explain in Section 2.1, is very different between various experiments. While for NC DIS at HERA heavy quark mass effects are very relevant and can be up to 20 % [31, 71] in some kinematic regime, for CC DIS at HERA there are practically irrelevant since Q^2 is large, see Table 2.1. However, when we consider CC DIS at small Q^2, as is present e.g. in neutrino experiments, they are again very relevant [72, 73].
- The mathematical structure in massless and massive diagrams is very different, e.g. the former yields collinear divergencies, see Sections 1.3 and 2.2.1, but the latter regulates them with explicit logarithms $\ln(Q^2/m^2)$, which makes them very different for interpolation grids, Section 2.2.3. Moreover, the massive diagrams contain far more special functions, which already makes the NLO diagrams no longer analytically solvable but only numerically tractable [3].
- Since the number of active flavors n_f is a direct dependency of the strong coupling $\alpha_s^{(n_f)}$ and PDFs $\mathbf{f}^{(n_f)}$ it appears explicitly in scale variations, Section 2.4. While for the renormalization scale variation it reduces to choosing the correct $\beta_0^{(n_f)}$ in Eq. (2.37), for the factorization scale variations it is more complicated as usual. First, it requires choosing the correct $\mathbf{P}^{(n_f)}$ in Eq. (2.54), but second it also requires the correct definition of the evolution basis. For example, now we can identify Eq. (2.60) as the four flavor singlet $f_\Sigma^{(4)}$, where as in three flavor we would get

$$f_\Sigma^{(3)}(x,Q^2) = f_d(x,Q^2) + f_{\bar{d}}(x,Q^2) + f_u(x,Q^2) + f_{\bar{u}}(x,Q^2) + f_s(x,Q^2) + f_{\bar{s}}(x,Q^2) \tag{2.95}$$

since, e.g., the charm is not an active quark and does not participate in Eq. (2.25).

2.6. Target mass corrections

In all our discussion so far we have implicitly assumed that the target, which often is the proton, see Section 2.1, is a massless particle. However, we can relax that condition and consider a target with finite mass M, which yields “Target Mass Corrections” (TMC). We refer to Refs. [74, 75] for a detailed review on the topic and instead focus here on the practical consequences.

Our master formula, Eq. (1.1), gives the so-called leading twist expression, which is dominant with respect to higher twist contributions, which are summarized as $\mathcal{O}\big(\Lambda_{\text{QCD}}^2/Q^2\big)$. While these higher twist terms are in principle not accessible in pQCD as PDFs are by definition leading order objects, they have recently gained renewed interest by PDF fitting groups [32–34]. TMCs are formally a part of these higher twist contributions, but since they can be computed systematically and consistently they are typically included in modern PDF extractions.

In practice, TMC amount to a shift in kinematics and a reweighting. We define

$$r = \sqrt{1 + \frac{4x^2M^2}{Q^2}} \tag{2.96}$$

with which we define the so-called Nachtmann variable [76]

$$\xi_N = \frac{2x}{1+r}\,. \tag{2.97}$$

Eventually, we find for the structure function F_2

$$F_2^{\text{TMC}}(x,Q^2) = \frac{x^2}{\xi_N^2 r^3} F_2^{\text{noTMC}}(\xi_N,Q^2) + \frac{6M^2x^3}{Q^2r^4} h_2(\xi_N,Q^2) + \frac{12M^4x^4}{Q^4r^5} g_2(\xi_N,Q^2) \tag{2.98}$$

where F_2^{noTMC} refers to the setup we have discussed so far and

$$h_2(\xi_N,Q^2) = \left(\tilde{h}_2 \otimes F_2^{\text{noTMC}}(Q^2)\right)(\xi_N) \qquad \text{with} \quad \tilde{h}_2(z) = \frac{z}{\xi_N}\,, \tag{2.99}$$

$$g_2(\xi_N,Q^2) = \left(\tilde{g}_2 \otimes F_2^{\text{noTMC}}(Q^2)\right)(\xi_N) \qquad \text{with} \quad \tilde{g}_2(z) = 1 - z\,. \tag{2.100}$$

For the other structure functions F_1, F_L, and F_3 one can find similar equations [74]. TMC are relevant for large x and/or small Q^2 as follows from Eqs. (2.96) and (2.97). Examining Eq. (2.98) we note that the structure function at the experimentally measured Bjorken-x corresponds to first approximation to a structure function computed at the Nachtmann variable ξ_N. This is a simple consequence from the reduced phase space, which is available for massive particles.

Note that Eqs. (2.99) and (2.100) are convolutions, i.e. they require the evaluation of $F_2^{\rm noTMC}$ at intermediate momentum fraction, which can be challenging in practice. Thus one often applies the following approximation [74]

$$F_2^{\rm TMC}(x,Q^2) \approx \frac{x^2}{\xi_N^2 r^3} F_2^{\rm noTMC}(\xi_N,Q^2)\left(1+\frac{6M^2 x\xi_N}{Q^2 r}(1-\xi_N)^2\right) \tag{2.101}$$

which only requires a single evaluation.

TMC introduce an explicit dependence on the target mass M into Eq. (1.1) and capture a consistent part of the higher twist contributions $\mathcal{O}\big(\Lambda_{\rm QCD}^2/Q^2\big)$. TMCs are independent from the requested perturbative accuracy or the exchanged boson since they refer to a kinematic correction and thus apply to every calculation. While they don't interfere directly with our previous content they must be applied separately on top of everything as a final operation after all other corrections have been taken into account.

2.7. PDF schemes

2.8. Summary

Fully inclusive DIS cross sections depend on

- the experimentally measured kinematic variables, Section 1.2, x, y, Q^2. They are not independent from each other, but linked through the invariant mass of the electron-proton system, Eq. (1.7), given by the experiment.
- the properties of the electro-weak bosons, Section 2.1. This includes basically all properties, such as the couplings to both leptons and quarks, specifically the electric charges and the electro-weak charges, or their respective masses.
- the discretization of interpolation grids, Section 2.2.3.
- the specific solution method for the RGE of the strong coupling and the PDFs, Section 2.3.
- the specific choice of renormalization and factorization scale as well as the applied factorization scale variation scheme, Section 2.4
- the heavy quark masses and the adopted FNS, Section 2.5.

- the target mass and the adopted approximation (if any), Section 2.6.

When performing a real-life PDF extraction all these dependencies must be taken into account and `yadism` [1, 2] provides one specific implementation thereof.

Chapter 3.

More advanced topics

3.1. γ_5

- helicity
- HVBM + Larin

3.2. Higher orders

- IHOU
- N3LO
- channels
- loops and legs

3.3. QED

- Qiu et al. [77–79]

3.4. Valo

- $C_\gamma \to k$

Appendix A.

Theory exercises

A.1. LO DIS

We are recomputing the LO DIS cross section depicted in Fig. A.1 in full generality. As we say in Section 1.2 we can decompose the full cross section into a leptonic tensor $L_{\mu\nu}$ and a hadronic tensor $W_{\mu\nu}$ where the red line indicates the split. In the collinear framework it is sufficient to compute the partonic (sub-)diagram $l + q \to l' + q' + X$. The parton-in-hadron amplitude ϕ_q combines upon taking the modulus square of the whole amplitude together with the inclusiveness X into the PDF $f_q(\xi)$. This leaves us essentially with twice Fig. 1.3a, once for the leptonic side and once for the quark side.

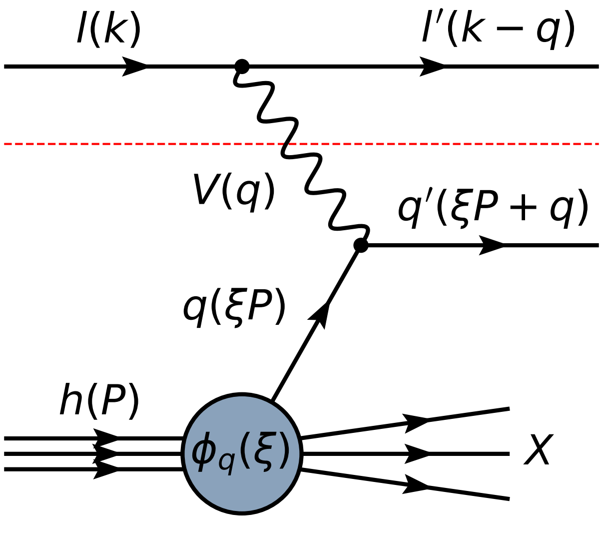

Figure A.1.: LO DIS Feynman diagram: $l + h \to l' + X$

Since the fermion lines are distinct we can compute them one at a time and if we keep them general enough we only need to do it once. Note that we first compute the lepton tensor, i.e. we keep the vector boson coupling Lorentz-index open and different in the amplitude and its complex conjugate.

In the first step we only deal with the Dirac structures. We need the following Feynman rules

- outgoing fermion $\bar{u}(k')$
- vector boson-fermion interaction Γ_μ, which we make concrete below
- incoming fermion $u(k)$

and we assume massless fermions in this exercise. Thus we have

$$\text{Fig. 1.3a} = \mathcal{A}_\mu = \bar{u}(k')\Gamma_\mu u(k) \tag{A.1}$$

where we have suppressed any additional dependency and we need to compute

$$T_{\mu\nu} = A_\mu A_\nu^* \,. \tag{A.2}$$

We recall some relevant Dirac traces:

$$\operatorname{tr}(\mathbb{1}) = 4 \tag{A.3}$$
$$\operatorname{tr}\big(\gamma_\mu\big) = 0 \tag{A.4}$$
$$\operatorname{tr}\big(\gamma_\mu\gamma_\nu\big) = 4g_{\mu\nu} \tag{A.5}$$
$$\operatorname{tr}\big(\gamma_\mu\gamma_\nu\gamma_\rho\big) = 0 \tag{A.6}$$
$$\operatorname{tr}\big(\gamma_\mu\gamma_\nu\gamma_\rho\gamma_\sigma\big) = 4(g_{\mu\nu}g_{\rho\sigma} - g_{\mu\rho}g_{\nu\sigma} + g_{\mu\sigma}g_{\nu\rho}) \tag{A.7}$$

We also need the simple anti-commutating γ_5, with properties

$$\gamma_5 = \frac{i}{4!}\varepsilon^{\mu\nu\rho\sigma}\gamma_\mu\gamma_\nu\gamma_\rho\gamma_\sigma, \quad \gamma_5\gamma_\mu = -\gamma_\mu\gamma_5, \quad \gamma_5\gamma_5 = \mathbb{1} \tag{A.8}$$

and the corresponding Dirac traces

$$\operatorname{tr}(\gamma_5) = \operatorname{tr}\big(\gamma_\mu\gamma_5\big) = \operatorname{tr}\big(\gamma_\mu\gamma_\nu\gamma_5\big) = \operatorname{tr}\big(\gamma_\mu\gamma_\nu\gamma_\rho\gamma_5\big) = 0 \tag{A.9}$$

and

$$\mathrm{tr}\left(\gamma_\mu \gamma_\nu \gamma_\rho \gamma_\sigma \gamma_5\right) = 4i\varepsilon_{\mu\nu\rho\sigma} \,. \tag{A.10}$$

We start by assuming $\Gamma_\mu = \gamma_\mu$ and averaging over fermion spins s upon which the spinors collapse to just, e.g., $\not{k}$.

Exercise A.1.1: Step 1

Proof

$$\begin{aligned} T^1_{\mu\nu} &= \frac{1}{2} \sum_s \bar{u}^{(s)}(k')\gamma_\mu u^{(s)}(k) \left(\bar{u}^{(s)}(k')\gamma_\nu u^{(s)}(k)\right)^* & \text{(A.11)} \\ &= 2\left(k_\mu k'_\nu + k_\nu k'_\mu - k \cdot k' g_{\mu\nu}\right) & \text{(A.12)} \end{aligned}$$

$T^1_{\mu\nu}$ becomes the spin-averaged leptonic tensor $L_{\mu\nu}$ for photons when choosing the lepton kinematics.

Next, we give up on averaging and instead consider fermions with definite helicities[1]. For spinors with a given helicity $\lambda = \pm 1$ we find

$$u(k,\lambda)\bar{u}(k,\lambda) = \frac{1}{2}\not{k}(1 - \lambda\gamma_5) \,. \tag{A.13}$$

Exercise A.1.2: Step 2

Proof

$$\begin{aligned} T^2_{\mu\nu} &= \bar{u}(k',\lambda)\gamma_\mu u(k,\lambda) \left(\bar{u}(k',\lambda)\gamma_\nu u(k,\lambda)\right)^* & \text{(A.14)} \\ &= T^1_{\mu\nu} + 2i\lambda\varepsilon_{\mu\nu\rho\sigma} k^\rho k'^\sigma & \text{(A.15)} \end{aligned}$$

$T^2_{\mu\nu}$ becomes the leptonic tensor $L^\gamma_{\mu\nu}$ for photons when choosing the lepton kinematics.

Next, we want to use a more realistic boson-fermion coupling Γ_μ, since eventually we need to consider the interference between a purely vectorial boson, such as the photon, and the Z boson, which has a mixture between a vectorial coupling and an axial vectorial coupling,

$$\Gamma^{ph}_\mu = g^{\gamma f}_V \gamma_\mu \,, \qquad \Gamma^Z_\mu = \gamma_\mu (g^{Zf}_V - g^{Zf}_A \gamma_5) \tag{A.16}$$

[1] Helicity is more convenient than spin in this context

where $g_V^{\gamma f}$ is the coupling strength between the photon and the fermion f and g_V^{Zf} and g_A^{Zf} are the vectorial and axial-vectorial coupling strength between the Z boson and the fermion f. By convention we exclude an explicit factor of the elementary electric charge from the definition of $g_{V,A}^{Bf}$. $T^2_{\mu\nu}$ obtains a trivial factor of $(g_V^{\gamma f})^2$ when using Γ^{ph}_μ.

Exercise A.1.3: Step 3

Proof

$$T^3_{\mu\nu} = \bar{u}(k',\lambda)\Gamma^Z_\mu u(k,\lambda)\left(\bar{u}(k',\lambda)\Gamma^{ph}_\nu u(k,\lambda)\right)^* \tag{A.17}$$
$$= g_V^{\gamma f}(g_V^{Zf} - \lambda g_A^{Zf})T^2_{\mu\nu} \tag{A.18}$$

$T^3_{\mu\nu}$ becomes the leptonic tensor $L^{\gamma Z}_{\mu\nu}$ for the photon-Z interference when choosing the lepton kinematics and couplings.

Finally, we need the pure Z boson contributions.

Exercise A.1.4: Step 4

Proof

$$T^4_{\mu\nu} = \bar{u}(k',\lambda)\Gamma^Z_\mu u(k,\lambda)\left(\bar{u}(k',\lambda)\Gamma^Z_\nu u(k,\lambda)\right)^* \tag{A.19}$$
$$= (g_V^{Zf} - \lambda g_A^{Zf})^2 T^2_{\mu\nu} \tag{A.20}$$

$T^4_{\mu\nu}$ becomes the leptonic tensor $L^Z_{\mu\nu}$ for the Z bosons when choosing the lepton kinematics and couplings.

The leptonic tensor $L^W_{\mu\nu}$ for the W bosons follows from the Z boson case by choosing appropriate couplings $g_{V,A}^{Wf}$.

The lepton kinematics are obtained by choosing the incoming momentum k and the the outgoing momentum $k' = k - q$, Fig. A.1.

Exercise A.1.5: Step 5: Ward identity

Proof

$$L^j_{\mu\nu} q^\mu = 0 = L^j_{\mu\nu} q^\nu \quad j \in \{\gamma, \gamma Z, Z, W\} \tag{A.21}$$

Exercise A.1.6: Step 6

Proof

$$L_{\mu\nu}^{\gamma} g^{\mu\nu} = -2Q^2 \tag{A.22}$$

We now turn our attention to the hadronic side, which is essentially identical except the fermion is now the quark and we have to choose different incoming and outgoing momenta. We define $p = \xi P$ and $T_{\mu\nu}^2$ becomes the photon-quark tensor $\hat{W}_{\mu\nu}^{\gamma q}$ by choosing the incoming momentum p and the outgoing momentum $p' = p + q$, Fig. A.1.

Exercise A.1.7: Step 7

Proof

$$L_{\mu\nu}^{\gamma} P^{\mu} P^{\nu} = \frac{Q^4}{x^2 y^2}(1-y) \tag{A.23}$$

Exercise A.1.8: Step 8

Proof

$$L_{\mu\nu}^{\gamma} \hat{W}_{\gamma q}^{\mu\nu} = 2\frac{Q^4}{z^2 y^2}\left(2 - 2y + y^2\right) \tag{A.24}$$

if we average over helicities.

A.2. RSL

Exercise A.2.1: RSL representation

Proof

$$C(z) = \left(\frac{\ln(1-z)}{1-z}\right)_+ \leftrightarrow C^R(z) = 0, C^S(z) = \frac{\ln(1-z)}{1-z}, C^L(x) = \frac{\ln(1-x)^2}{2} \tag{A.25}$$

Exercise A.2.2: Product of functions 1

Give the RSL representation of the product a regular function $r(z)$ and a $+$-distribution $(p(z))_+$, $C(z) = r(z)(p(z))_+$

Exercise A.2.3: Product of functions 2

Proof

$$[r(z)p(z)]_+ = r(z)\,[p(z)]_+ - \delta(1-z)\int\limits_0^1 dy\; r(y)\,[p(y)]_+ \tag{A.26}$$

where again $r(z)$ is a regular function and $(p(z))_+$ is a $+$-distribution.

A.3. Resummation

Exercise A.3.1: LO α_s evolution

Proof Eq. (2.21) can be written as

$$\alpha_s(\mu_R^2) = \frac{1}{\beta_0 \ln\left(\mu_R^2/\Lambda_{\mathrm{QCD}}^2\right)} \tag{A.27}$$

and given an explicit expression for Λ_{QCD}.

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